

**MINDPULSE: A MULTI-MODAL AI SYSTEM  
FOR EARLY DETECTION OF MENTAL  
HEALTH CRISIS IN YOUNG ADULTS**

**A PROJECT REPORT**

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# TABLE OF CONTENTS

|                                                                           |        |
|---------------------------------------------------------------------------|--------|
| Abstract                                                                  | vii    |
| List of Tables                                                            | viii   |
| List of Figures                                                           | ix     |
| List of Symbols and Abbreviations                                         | x      |
| <br>1 INTRODUCTION                                                        | <br>1  |
| 1.1 General                                                               | 1      |
| 1.2 Purpose                                                               | 3      |
| 1.3 Scope                                                                 | 4      |
| 1.4 Artificial Intelligence and Machine Learning                          | 5      |
| 1.5 Explainable Artificial Intelligence (XAI)                             | 9      |
| 1.6 Large Language Models (LLMs)                                          | 9      |
| <br>2 LITERATURE REVIEW                                                   | <br>11 |
| 2.1 Mental Health and Digital Phenotyping                                 | 11     |
| 2.2 Machine Learning for Mental Health Prediction                         | 12     |
| 2.3 Explainable Artificial Intelligence in Health Care                    | 13     |
| 2.4 Temporal Modeling for Longitudinal Mental Health Analysis             | 14     |
| 2.5 Large Language Models in Mental Healthcare                            | 15     |
| 2.6 Class Imbalance in Mental Health Data                                 | 16     |
| 2.7 Clinical Evidence for Differential Weighing of Depression and Anxiety | 16     |
| 2.8 Deployment Frameworks for Real-Time Mental Health Systems             | 17     |
| 2.9 Research Gap and Motivation                                           | 18     |
| <br>3 PROPOSED METHODOLOGY                                                | <br>19 |
| 3.1 Overview of Proposed System                                           | 19     |
| 3.2 Dataset Description                                                   | 20     |
| 3.3 Data Preprocessing                                                    | 24     |
| 3.4 Machine Learning Model Development                                    | 26     |
| 3.5 Model Evaluation Metrics                                              | 28     |
| 3.6 Mental Health Index (MHI)                                             | 29     |
| 3.7 Risk Escalation Engine (REE)                                          | 31     |
| 3.8 SHAP-Explainable AI Integration                                       | 35     |
| 3.9 Temporal Modeling for Longitudinal Trend Analysis                     | 35     |
| 3.10 Personalized AI Recommendations                                      | 38     |

|                                                         |    |
|---------------------------------------------------------|----|
| 4 RESULTS                                               | 39 |
| 4.1 Experimental Setup                                  | 40 |
| 4.2 Performance Evaluation Metrics                      | 41 |
| 4.3 Model Performance Comparison                        | 41 |
| 4.4 Mental Health Index Analysis                        | 42 |
| 4.5 Risk Escalation Engine Evaluation                   | 43 |
| 4.6 Explainable AI Outcomes                             | 44 |
| 4.7 Temporal Analysis for Longitudinal Trend Monitoring | 45 |
| 4.8 Personalized AI Recommendations                     | 47 |
| 5 CONCLUSION                                            | 48 |
| 6 FUTURE SCOPE                                          | 50 |
| 7 REFERENCES                                            | 52 |
| 8 APPENDIX A                                            | 54 |
| 9 APPENDIX B                                            | 57 |
| 10 APPENDIX C                                           | 59 |

## ABSTRACT

Rising rates of psychological distress among young adults have created urgent demand for proactive, technology-enabled screening tools. This work presents MindPulse, an integrated AI platform combining supervised machine learning, Shapley Additive Explanations (SHAP) for interpretability, and a large language model (LLM) to deliver end-to-end mental health crisis support. Training and evaluation were performed on a clinical dataset of 15,147 assessments spanning 47 features. Three classifiers, namely Random Forest, Logistic Regression, and XGBoost, were evaluated. Random Forest attained the best overall performance. Risk quantification is managed by a novel Risk Escalation Engine (REE) that fuses ML severity predictions, standardised clinical thresholds, a composite Mental Health Index (MHI), early-warning signals, and PHQ-9 safety flags into a single 0–7 escalation score. SHAP values make every prediction auditable by attributing model output to individual symptom drivers. A longitudinal tracking module captures trajectory changes across repeated assessment sessions, while the Llama-3.3-70B-Versatile LLM generates context-specific coping strategies and crisis resources. The full pipeline is deployed via FastAPI and React, supporting live inference, visual dashboards, and automated clinical alerts. Validation results confirm the system's accuracy and its capacity to produce clinically meaningful, transparent risk assessments, establishing MindPulse as a scalable foundation for AI-assisted digital mental healthcare.

## LIST OF TABLES

|     |                                                            |    |
|-----|------------------------------------------------------------|----|
| 3.1 | Complete Dataset Information                               | 21 |
| 3.2 | Complete Dataset Feature Description                       | 22 |
| 3.3 | MHI Wellness Tiers and Clinical Interpretation             | 31 |
| 3.4 | MHI Severity-to-Risk Conversion Functions                  | 33 |
| 3.5 | Risk Escalation Thresholds and Clinical Protocols          | 35 |
| 3.6 | Trend Classification Thresholds                            | 36 |
| 3.7 | Volatility Classification Thresholds                       | 37 |
| 4.1 | Implementation Environment                                 | 39 |
| 4.2 | Experimental Configuration Summary                         | 40 |
| 4.3 | Comparative Analysis of Machine Learning Model Performance | 41 |
| 4.4 | Training and Testing Accuracies of Machine Learning Models | 42 |

## LIST OF FIGURES

|     |                                                                  |    |
|-----|------------------------------------------------------------------|----|
| 1.1 | Logistic Regression as a Baseline Classifier                     | 7  |
| 1.2 | Schematic Diagram of Random Forest Model                         | 8  |
| 1.3 | Mathematics Behind XGBoost Algorithm                             | 8  |
| 1.4 | SHAP Diagram                                                     | 9  |
| 3.1 | System Architecture Diagram                                      | 20 |
| 3.2 | Data Preprocessing Workflow Diagram                              | 25 |
| 3.3 | Class Distribution: (a) Before Using SMOTE (b) After Using SMOTE | 26 |
| 3.4 | Machine Learning Pipeline Diagram                                | 27 |
| 3.5 | Risk Escalation Engine Block Diagram                             | 32 |
| 4.1 | MHI Score Gauge Visualization                                    | 43 |
| 4.2 | Risk Escalation Score                                            | 44 |
| 4.3 | SHAP XAI Results                                                 | 44 |
| 4.4 | Temporal Analysis for Longitudinal Trend Monitoring              | 46 |
| 4.5 | Personalized AI Recommendations                                  | 47 |

## LIST OF SYMBOLS AND ABBREVIATIONS

|              |                                            |
|--------------|--------------------------------------------|
| <b>AI</b>    | Artificial Intelligence                    |
| <b>API</b>   | Application Programming Interface          |
| <b>CSS</b>   | Cascading Style Sheets                     |
| <b>D</b>     | Distress Index                             |
| <b>E</b>     | Escalation Score                           |
| <b>GAD-7</b> | Generalized Anxiety Disorder Assessment-7  |
| <b>LLM</b>   | Large Language Model                       |
| <b>MHI</b>   | Mental Health Index                        |
| <b>ML</b>    | Machine Learning                           |
| <b>N</b>     | Number of Assessment Sessions              |
| <b>PHQ-9</b> | Patient Health Questionnaire-9             |
| <b>REE</b>   | Risk Escalation Engine                     |
| <b>SHAP</b>  | Shapley Additive Explanations              |
| <b>SMOTE</b> | Synthetic Minority Over-sampling Technique |
| <b>XAI</b>   | Explainable Artificial Intelligence        |
| $\Delta$     | Delta (Change)                             |
| $\sigma$     | Standard Deviation                         |

# CHAPTER 1

## INTRODUCTION

### 1.1. General

Psychological well-being underpins every aspect of human functioning, shaping cognition, emotion, and social engagement alike. The World Health Organization ranks mental disorders among the leading global causes of disability, with depressive illness alone estimated to affect more than 280 million people [1]. The ripple effects extend well beyond affected individuals; families, institutions, and economies bear substantial secondary burdens through productivity losses, increased healthcare utilisation, and diminished life quality across communities.

Among all age groups, those between 18 and 30 carry a disproportionate mental health burden. This period demands simultaneous navigation of high-stakes transitions, including leaving adolescence, entering tertiary education or employment, achieving financial independence, and building lasting social bonds. Each of these transitions is individually stressful; experienced concurrently, they create compounding psychological pressure. Contemporary stressors compound the challenge further: competitive academic environments, economic uncertainty, pervasive social-media comparison culture, and chronic digital over-exposure collectively elevate vulnerability in this cohort [1], [2].

Epidemiological data from the National Institute of Mental Health confirm that young adults record the highest documented rates of anxiety disorders and major depressive episodes of any demographic group [2]. The convergence of developmental sensitivity with elevated environmental stress creates a critical window: timely identification and intervention during this period can substantially alter the long-term trajectory of psychological illness. Absent adequate and timely support, sub-threshold psychological symptoms can escalate into diagnosable clinical conditions, with downstream consequences including academic disengagement, fractured relationships, substance misuse, self-harm, and, in the most severe cases, suicidal behaviour.

The critical role of early detection in addressing mental health crises among young adults is well-established, yet numerous obstacles continue to hinder timely identification and intervention. A key obstacle is the persistent stigma attached to mental illness, which prevents many individuals from disclosing symptoms or seeking professional support. Compounding this, mental health services remain inaccessible in resource-limited environments. Even in well-resourced systems, conventional assessments are typically conducted at discrete intervals, creating gaps during which deterioration can go undetected.

Conventional mental health evaluation relies on structured clinical interviews, standardized questionnaires, and expert-administered assessments. While these tools have proven reliable across populations, they carry inherent limitations: they demand scarce professional time and expertise, capture only a momentary impression of a person's psychological state rather than tracking it continuously, and depend entirely on accurate self-report, which can be undermined by social desirability bias, limited emotional self-awareness, and poor insight into one's condition.

The PHQ-9 for depression and the GAD-7 for anxiety are two of the most commonly utilized screening instruments in mental health practice. The PHQ-9 maps directly to the DSM diagnostic criteria for major depressive disorder, covering nine symptom domains including mood, motivation, sleep quality, energy, appetite, self-worth, concentration, psychomotor function, and suicidal ideation [13], [14].

The GAD-7 captures seven anxiety-related domains including worry, restlessness, irritability, and fear of adverse outcomes [15], [16]. While both instruments provide clinically meaningful screening data, their utility is bounded by the same constraints inherent to all self-report measures when applied in isolation.

The growing integration of AI and ML in healthcare has opened new possibilities for mental health risk detection through data-driven analysis of complex, multi-source inputs. These approaches augment conventional screening by enabling continuous monitoring, automated risk stratification, and personalized guidance. ML models trained on questionnaire responses,

behavioral signals, and contextual variables can surface risk patterns that traditional clinical assessments may miss [3]–[5].

A key challenge in applying ML to mental health is the interpretability gap: sophisticated models like gradient boosting ensembles deliver risk scores without transparent clinical reasoning, undermining trust among both clinicians and patients [6], [7]. Additionally, most existing systems focus solely on prediction without providing actionable, personalized guidance tailored to individual risk profiles.

## **1.2. Purpose**

This project constructs a practical, scalable, and clinically grounded platform for early-stage mental health crisis identification in young adults. The central objective is the realisation of MindPulse, a multimodal AI-powered early warning system, that unifies several complementary artificial intelligence methods to achieve comprehensive risk assessment, transparent model interpretation, and individually tailored clinical guidance.

Six concrete objectives drive the project. First, construct and rigorously benchmark three supervised classifiers, namely Logistic Regression, Random Forest, and XGBoost, on standardised assessment data, identifying which algorithm best balances predictive accuracy, computational efficiency, and interpretability for live deployment.

Second, embed SHAP-based explainability to demystify model outputs. The opacity of complex ML algorithms has historically impeded healthcare adoption; surfacing symptom-level prediction drivers enables clinicians and patients to audit and appropriately trust system recommendations.

Third, develop the novel Mental Health Index (MHI), a composite metric that normalises and blends PHQ-9 and GAD-7 scores onto a clinician-readable 0–100 wellness continuum, facilitating intuitive monitoring and longitudinal comparison of psychological status.

Fourth, engineer a multi-signal Risk Escalation Engine (REE) that synthesises ML severity

outputs, raw clinical scores, MHI wellness bands, early-warning triggers, and PHQ-9 safety flags into a single actionable escalation score, moving beyond binary classification toward nuanced, intervention-ready risk stratification.

Fifth, implement a temporal modelling module to track MHI trajectories, detect deterioration or recovery patterns, quantify risk persistence, and measure wellness volatility across repeated assessment sessions. track changes in mental health status across multiple assessment sessions and detect patterns of deterioration or improvement.

Sixth, integrate the Llama-3.3-70B-Versatile LLM to translate structured risk profiles into personalised, plain-language coping strategies and crisis resources, thereby closing the loop between risk detection and practical support.

Finally, deploy the complete pipeline within a production-grade web architecture (FastAPI backend, React 18 frontend) to deliver real-time inference, interactive visualisations, and automated crisis alerts, thereby demonstrating operational feasibility at scale.

### **1.3. Scope**

This thesis covers the end-to-end conception, construction, and evaluation of the MindPulse multimodal AI framework. The system targets the 18–30-year-old demographic, reflecting both the training dataset composition and the epidemiologically established vulnerability of this population [1], [2].

Input data derive from standardised PHQ-9 and GAD-7 assessments, augmented by supplementary indicators covering mood, behaviour, stress, sleep quality, and emotional state. The complete dataset, comprising 15,147 records across 47 features, was sourced from the Hugging Face platform.

The project spans six domains: ML model development and evaluation; XAI integration; MHI construction; REE design; longitudinal temporal modelling; LLM recommendation generation; and full-stack web deployment. large

language models for generating personalized recommendations; and the deployment of a full-stack web application for real-time system operation.

Several areas fall outside the current scope. Physiological wearable data and social-media behavioural signals are excluded despite their recognised value. Critically, MindPulse is a screening and decision-support tool, not a diagnostic instrument, and it augments rather than replaces professional clinical judgement. The dataset is restricted to English-language responses, precluding cross-cultural or multilingual validation.

Target deployment contexts include university wellbeing services, corporate employee assistance programmes, and digital health platforms conducting routine psychological screening. MindPulse serves dual roles: a clinical decision-support layer for healthcare professionals, and a self-monitoring interface enabling individuals to track their own mental health longitudinally.

## **1.4. Artificial Intelligence, Machine Learning and Explainable AI**

### **1.4.1 Artificial Intelligence**

Artificial intelligence (AI) encompasses the design of computational systems that replicate cognitive functions such as perception, reasoning, and autonomous decision-making. Modern AI is predominantly task-specific (narrow AI), with demonstrated capabilities exceeding human efficacy in domains such as image recognition and linguistic processing, though general-purpose AI, which would be capable of performing any intellectual task flexibly, remains a long-term research objective.

Healthcare has become a primary AI application domain, with deployments spanning radiological image analysis, computational drug discovery, continuous patient surveillance, and clinical outcome prediction. Mental healthcare is especially receptive: psychological conditions produce rich, heterogeneous data signals (questionnaire responses, behavioural patterns, and contextual variables) that exceed the synthesising capacity of conventional clinical assessment, making AI augmentation particularly valuable.

### **1.4.2 Machine Learning**

Machine learning (ML) is a subset of artificial intelligence (AI) centered on building algorithms that improve through experience rather than explicit instruction. These algorithms identify statistical patterns and structural regularities within data, enabling generalization to previously unseen inputs.

ML is broadly divided into supervised learning (used here for risk classification), unsupervised learning, and reinforcement learning.

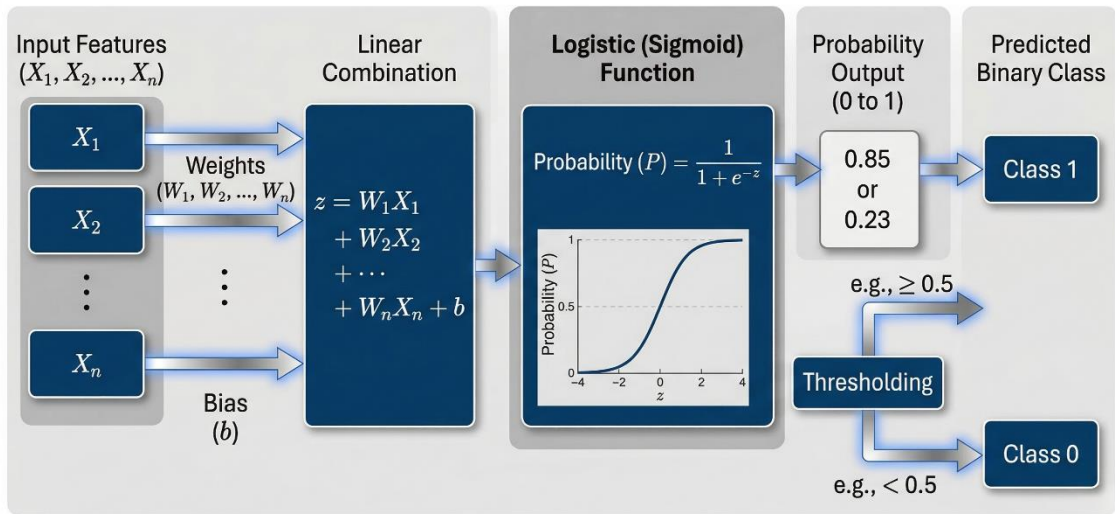
Unsupervised learning uncovers patterns in unlabeled data. Typical unsupervised learning problems encompass clustering, dimensionality reduction, and anomaly identification.

Reinforcement learning optimizes behavior via reward signals. For this study, supervised learning is the most relevant paradigm. Ensemble methods deliver consistently strong performance on structured health datasets [4], [5].

### **1.4.3 Linear Models**

Linear models express outputs as weighted sums of input features, balancing interpretability with computational efficiency.

Logistic Regression is the primary linear model in this study, adapting the linear framework for classification by applying a sigmoid transformation to produce class probabilities. It serves as a transparent, interpretable baseline against which ensemble models are benchmarked, with threshold-based decisions as shown in Fig. 1.1.

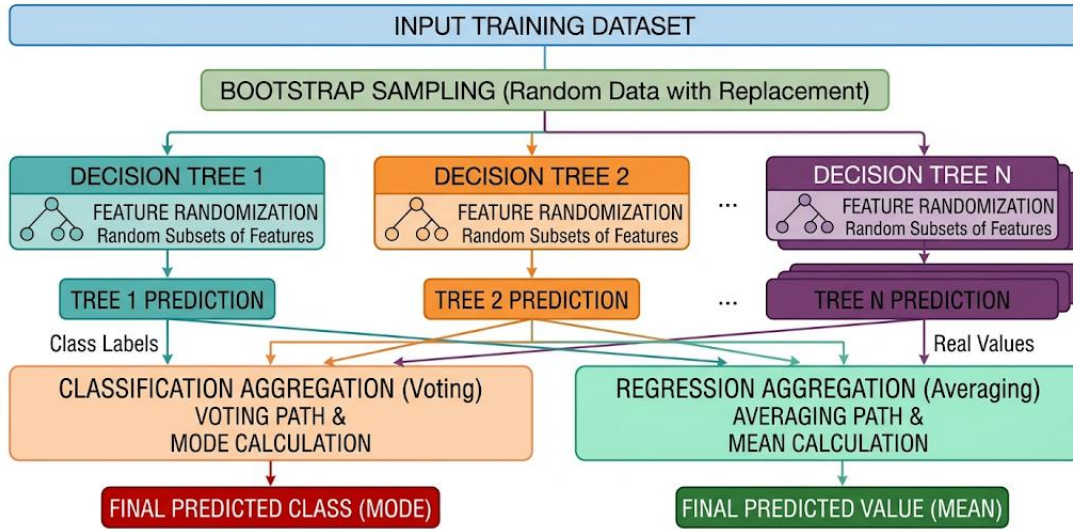


**Figure 1.1. Logistic Regression as a Baseline Classifier**

#### 1.4.4 Ensemble Learning Methods

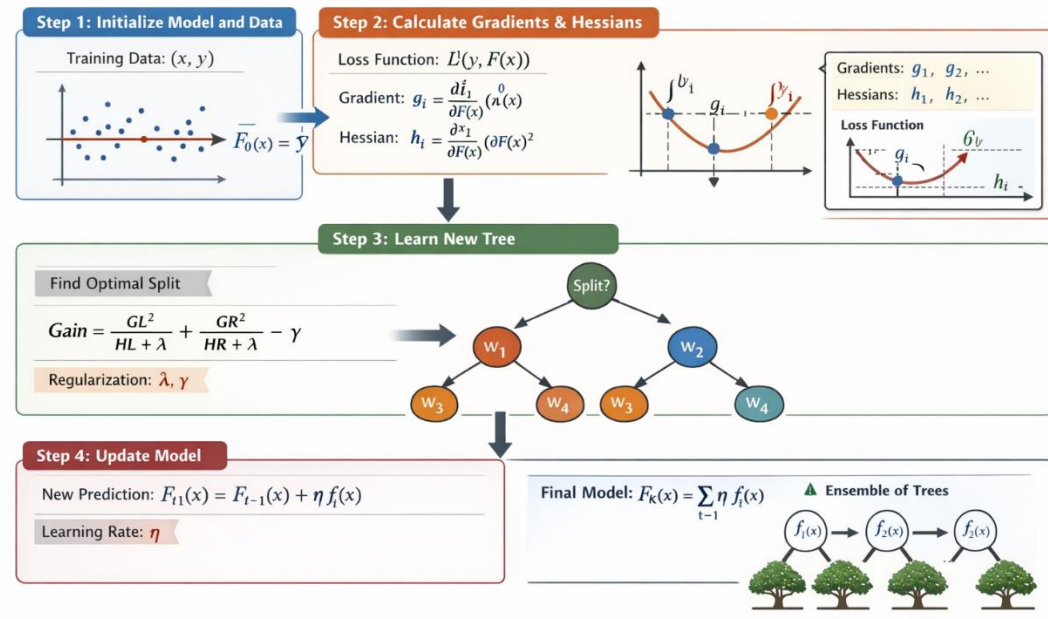
Ensemble methods pool predictions from multiple base learners; because individual model errors tend to cancel upon aggregation, the combined output is more accurate and stable. This study employs Random Forest and XGBoost as its two ensemble approaches.

Random Forest trains many decision trees, each on a bootstrapped data sample with random feature subsampling at every split. Majority voting across trees (classification) or averaging (regression) produces the final output. This dual randomization decorrelates trees, controls overfitting, and yields a model that is robust to noisy, high-dimensional inputs, properties well-suited to health data [4], [5].



**Figure 1.2. Schematic Diagram of Random Forest Model**

XGBoost is a high-performance gradient boosting implementation that constructs decision trees sequentially, with each subsequent tree correcting the remaining flaws of the preceding ensemble. It incorporates L1/L2 regularization, robust missing-value handling, and parallel computation to maximize both accuracy and efficiency (Fig. 1.3). In mental health classification tasks, XGBoost performs comparably to Random Forest on structured assessment data [4], [5].

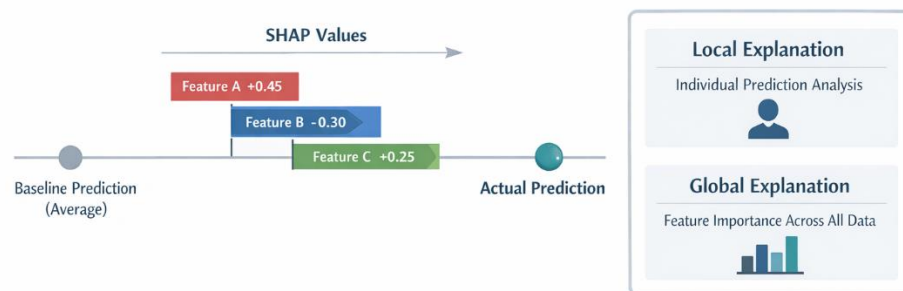


**Figure 1.3. Mathematics Behind XGBoost Algorithm**

## 1.5. Explainable Artificial Intelligence (XAI)

As ML models grow in complexity and permeate high-stakes domains, the demand for intelligible decision-making has intensified. XAI methods produce explanations of individual predictions or aggregate model behaviour that enable clinicians, patients, and regulators to understand, validate, and appropriately calibrate trust in AI-generated outputs. In clinical settings specifically, both providers and patients require clear justifications for AI risk assessments before acting on them [6], [7].

SHAP (Shapley Additive Explanations) is a widely adopted interpretability framework built around cooperative game theory. It disaggregates each prediction into contributions from individual features, relative to a mean baseline, quantifying how much each input shifted the output from the average expectation. SHAP offers both local and global explanations, elucidating individual predictions and summarizing feature importance across all predictions, respectively. Local explanations are particularly beneficial in clinical settings as they expose which specific symptoms drove a particular individual's risk score (Fig. 1.4). At the same time, global summaries validate that the model relies on clinically relevant features across the full population [6], [7]. Embedding SHAP into MindPulse directly addresses the interpretability gap common to complex ensemble models.



**Figure 1.4. SHAP Diagram**

## 1.6 Large Language Models (LLMs)

Large language models are generative deep neural networks that have undergone pre-training on vast text corpora, conferring the ability to produce fluent, contextually coherent natural

language across diverse tasks. Their demonstrated strengths in comprehension, synthesis, and reasoning have attracted growing healthcare interest, with applications ranging from automated clinical documentation and patient communication to health information summarisation [8], [9].

In mental healthcare specifically, LLMs offer the prospect of converting structured clinical data into nuanced, person-centred guidance that rule-based systems cannot deliver. MindPulse leverages Llama-3.3-70B-Versatile model, a 70-billion-parameter instruction-tuned architecture from Meta (accessed via Groq [10] and documented on Hugging Face [11]), as its recommendation engine. The model receives a structured prompt encoding the user's risk tier, MHI score, and top SHAP-ranked symptom contributors, generating four output categories: targeted coping strategies, wellness maintenance recommendations, risk-calibrated preventive guidance, and crisis resources for high-escalation cases.

## CHAPTER 2

### LITERATURE REVIEW

This chapter situates MindPulse within its scholarly context by reviewing published evidence across the six technical domains the system integrates: standardised mental health assessment and digital phenotyping; ML-based risk prediction; XAI in clinical settings; longitudinal modelling of psychological trajectories; LLM applications in mental healthcare; and class imbalance remediation. It concludes by identifying the specific research gap the framework addresses and articulating the motivation for its integrated design.

#### 2.1. Mental Health Assessment and Digital Phenotyping

The predictive validity of any data-driven mental health system rests directly on the quality of its assessment inputs. Two instruments dominate clinical and research screening for depression and anxiety: the Patient Health Questionnaire-9 and the Generalized Anxiety Disorder-7. Carroll et al. [13] reviewed the PHQ-9's psychometric performance across diverse clinical contexts, confirming its reliability and validity. The nine-item tool maps directly to DSM diagnostic criteria for major depressive disorder, with each item scored 0–3. Total scores range from 0–27, with validated severity bands from minimal to severe depression, and the review also noted opportunities to improve its clinical application.

Fonseca-Pedrero et al. [14] confirmed the PHQ-9's factorial validity and cross-group measurement invariance, explicitly endorsing its suitability for the young adult population that MindPulse targets. The stability of item structure across demographic subgroups makes the instrument a reliable feature source for ML-based predictive modelling.

Zhang et al. [15] validated the GAD-7 across diverse cohorts, confirming its internal consistency, test-retest reliability, and construct validity. Scores vary from 0 to 21, with

Thresholds of 5, 10, and 15 correspond to mild, moderate, and severe anxiety, respectively. The GAD-7 utilizes a 0–3 scoring system, yielding a total score range of 0 to 21, with cut-off thresholds at 5 (mild), 10 (moderate), and 15 (severe) anxiety. The study furnished evidence supporting the instrument's internal consistency, test-retest reliability, and concept validity, thereby affirming its appropriateness for anxiety screening across diverse populations.

Vrublevska et al. [16] independently replicated these findings in a separate cohort, confirming the GAD-7's cross-population stability. Taken together, the four validation studies [13]–[16] establish both instruments as psychometrically rigorous tools well-suited as ML model inputs.

Beyond the traditional questionnaire-based assessment, the concept of digital phenotyping has emerged as a complementary approach to mental health monitoring. Bufano et al. [3] investigated digital phenotyping for the surveillance of mental disorders, analyzing how passively collected device traces and environmental indicators can reliably infer mental health status, compensating for gaps between episodic assessments. This evidence supports MindPulse's approach of integrating standardized assessment instruments with additional behavioral and contextual indicators to create a comprehensive picture of an individual's mental health status.

## **2.2. Machine Learning for Mental Health Prediction**

Data-driven prediction of mental health outcomes has attracted sustained research investment, propelled by growing dataset availability and advances in ML methodology. This section reviews studies most directly relevant to the algorithmic and evaluation choices made within MindPulse.

Hasan et al. [4] systematically reviewed supervised ML approaches to mental health detection, covering decision trees, SVMs, neural networks, and ensemble models on structured assessment data. Their analysis highlighted that algorithm selection, feature engineering, and evaluation design are all critical to model quality, and identified ensemble methods as consistently top-performing on mental health classification benchmarks.

Trelles et al. [5] conducted a meta-analysis of supervised classifiers for depression detection.

Pooling evidence across multiple studies and datasets, they found that ensemble approaches, particularly Random Forest and gradient boosting, outperformed traditional methods like logistic regression and SVMs, providing quantitative justification for the model selection strategy in this study.

Evidence from Hasan et al. [4] and Trelles et al. [5] collectively validates the multi-algorithm benchmarking strategy employed in MindPulse and the ultimate selection of Random Forest for deployment. Literature consensus identifies this algorithm's favourable balance of accuracy, noise robustness, and compatibility with post-hoc explainability as optimal for clinical-facing applications.

However, an important limitation identified across the ML-for-mental-health literature is the lack of model interpretability. Both Hasan et al. and Trelles et al. noted that while ensemble methods achieve high predictive accuracy, their complex internal structure makes it difficult for clinicians and end-users to understand how specific input features contribute to individual predictions. This limitation, commonly referred to as the "black-box problem" restricts the clinical utility of ML predictions because providers cannot validate their reasoning. This interpretability gap directly motivates SHAP integration as a primary MindPulse component.

### **2.3. Explainable Artificial Intelligence in Healthcare**

The proliferation of complex ML systems in consequential settings has established XAI as a research priority in its own right. XAI methods produce explanations of individual predictions or aggregate model behaviour, enabling clinicians, patients, and auditors to understand, validate, and calibrate trust in AI-generated outputs.

Mienye et al. [6] surveyed XAI applications in healthcare, organizing techniques into model-agnostic and model-specific categories. SHAP was highlighted among model-agnostic methods for its game-theoretic guarantees of uniqueness, consistency, and efficiency in feature attribution. The survey established that clinicians, patients, and regulators each have distinct but equally pressing needs for transparency in AI-assisted medical decisions.

Chaddad et al. [7] compared SHAP, LIME, attention, and gradient-based XAI methods across

healthcare tasks. Their analysis involved a comparison of the strengths and limitations of various XAI techniques across healthcare tasks, finding SHAP to combine the strongest theoretical foundations with dual instance-level and population-level explanatory power. This helps clinicians identify which symptoms drive individual risk assessments and validate model-wide behavior.

Evidence from both surveys [6], [7] validates SHAP as the most principled choice for MindPulse's explainability layer. Its per-feature, per-prediction attribution capability directly addresses the clinical need: practitioners can identify exactly which symptom contributions drove a specific risk score, supporting transparent, auditable assessment decisions.

#### **2.4. Temporal Modeling for Longitudinal Mental Health Analysis**

Psychological conditions are inherently dynamic; they evolve through episodes of deterioration, stabilisation, and recovery that cross-sectional assessments cannot adequately capture. This limitation has motivated research into longitudinal modelling approaches that analyse health trajectories across repeated measurement occasions.

Bufano et al. [3] investigated digital phenotyping methodologies that inherently integrate the temporal dimensions of mental health data. Their research on the surveillance of mental disorders through digital phenotyping illustrated that monitoring behavioral and psychological indicators over time facilitates the identification of changes and trends that may not be discernible from isolated assessments. The temporal aspect of digital phenotyping underpins the continuous monitoring strategy employed in the MindPulse system.

Pujadas et al. [12] demonstrated that treating sequential assessment data as a temporal signal substantially improves detection of youth at risk of mental health deterioration. Their ML-based longitudinal analysis showed that change patterns across assessments carry predictive information systematically overlooked by cross-sectional models, providing direct justification for MindPulse's session-based tracking architecture.

Building on both Bufano et al. [3] and Pujadas et al. [12], MindPulse's temporal module

computes five complementary longitudinal metrics, namely global MHI trend ( $\Delta\text{MHI}$ ), recent deterioration flags, risk-level persistence streaks, wellness volatility ( $\sigma_M$ ), and trajectory direction, to characterise the dynamic arc of each user's psychological health across successive assessments.

## 2.5. Large Language Models in Mental Healthcare

Advances in large language modelling have opened new possibilities for personalised digital mental health support, particularly the generation of tailored recommendations that adapt to individual clinical profiles rather than issuing generic advice. LLMs trained on broad text corpora demonstrate strong capacity for producing coherent, contextually appropriate responses across health and wellness domains.

Lawrence et al. [8] weighed the advantages and hazards of large language models in mental health, providing a balanced assessment of the potential benefits and concerns associated with deploying LLMs in this sensitive domain. On the opportunity side, the authors discussed the potential for LLMs to provide accessible, personalized mental health support, including psychoeducation, coping strategy recommendations, and crisis guidance. The ability of LLMs to generate natural language responses based on structured input data (such as predicted risk levels and symptom profiles) enables a form of personalized intervention that can complement traditional clinical care. On the risk side, the authors highlighted concerns about the accuracy and appropriateness of LLM-generated mental health advice, the potential for harmful outputs in crisis situations, and the importance of establishing appropriate safety guardrails.

Jin et al. [9] catalogued expanding LLM applications in mental health, including automated screening, supportive dialogue agents, sentiment analysis, treatment recommendation, and psychoeducational content generation, while emphasising that responsible implementation demands rigorous attention to output accuracy, safety protocols, and ethical constraints. These considerations inform MindPulse's constrained, structured-prompt approach to LLM integration.

MindPulse employs Llama-3.3-70B-Versatile model, a 70-billion-parameter Llama-family

instruction-tuned architecture accessible via Groq [10] and documented on Hugging Face [11], as its recommendation engine. The model converts structured prompts encoding risk level, MHI score, and top SHAP contributors into personalised guidance across four output categories.

## **2.6. Class Imbalance in Mental Health Data**

Class imbalance is a structural feature of mental health screening datasets: low-risk individuals predominate while clinically critical high-risk cases are rare. Left unaddressed, this asymmetry biases classifiers toward majority-class prediction, suppressing sensitivity precisely where it matters most. This disparity can skew machine learning models towards forecasting the majority class, thereby leading to reduced sensitivity in recognizing high-risk individuals.

Chen et al. [17] surveyed the imbalanced learning landscape, comparing oversampling, undersampling, cost-sensitive, and ensemble-based approaches. SMOTE emerged as the most widely validated oversampling technique: rather than duplicating minority instances, it synthesises new observations through linear interpolation between k-nearest neighbours, enriching minority-class representation while preserving distributional characteristics.

Consistent with these recommendations [17], MindPulse applies SMOTE before training to equalise risk-category frequencies, countering majority-class bias and improving sensitivity to high-risk cases where missed detections carry the greatest clinical cost.

## **2.7 Clinical Evidence for Differential Weighting of Depression and Anxiety**

The MHI's 0.6/0.4 weighting of depression over anxiety reflects an evidence-based clinical judgement about their relative functional impact. The studies below substantiate this asymmetry.

Kim et al. [18] compared the functional burden of depression and anxiety in comorbid presentations, finding that depressive symptoms were more strongly associated with impaired

daily functioning, occupational performance, and quality of life. This differential impact provides empirical grounding for the MHI's higher depression coefficient.

Zhou et al. [19] produced convergent evidence: depressive symptoms showed stronger statistical associations with disability and quality-of-life impairment than anxiety symptoms, independently corroborating Kim et al.'s findings and reinforcing the higher depression weight in the MHI formula.

Arnone et al. [20] meta-analytically documented a robust association between depressive severity and suicidal behaviour. This provides additional justification for prioritising depression signals in the REE and for treating PHQ-9 Item 9 (suicidal ideation) as a standalone high-priority safety flag.

Yarborough et al. [21] demonstrated prospectively that elevated PHQ-9 Item 9 scores predict subsequent suicidal behaviour, validating the REE design decision to treat this item as a standalone safety flag capable of triggering maximum-priority escalation independently of the overall severity score.

## **2.8 Deployment Frameworks for Real-Time Mental Health Systems**

Deploying ML-based mental health tools in production requires frameworks that support low-latency inference, rich data visualisation, and responsive user interfaces. The FastAPI–React combination has become a standard production stack for serving ML models in real-time applications.

FastAPI's asynchronous request handling, Pydantic schema validation, and automatic OpenAPI documentation make it well-suited for prediction APIs requiring concurrency, enabling efficient handling of concurrent requests, making it suitable for real-time prediction applications. React, developed by Meta, is a JavaScript library for building user interfaces, providing a component-based architecture that enables interactive, responsive web applications. The combination of these technologies supports the development of scalable platforms that can deliver real-time predictions, visualizations, and alerts to end users.

Existing web-deployed mental health platforms predominantly focus on prediction delivery, rarely consolidating explainability, longitudinal tracking, and LLM-driven guidance within a single coherent interface. This gap underscores MindPulse's architectural contribution.

## **2.9 Research Gap and Motivation**

The literature review reveals that while each of MindPulse's constituent technologies has been studied independently, no published system has integrated them within a single clinically coherent framework. Four specific gaps emerge: (1) most mental health ML models lack embedded interpretability, restricting clinical adoption [4]–[7];

(2) temporal longitudinal analysis has not been systematically coupled with real-time ML prediction and XAI in deployed systems [3], [12]; (3) LLM-generated guidance has not been integrated with ML risk scores, SHAP explanations, and session-history tracking within a unified platform [8], [9];

The exploration of multi-signal composite risk engines fusing ML predictions, clinical scores, wellness indices, and safety flags remains understudied. MindPulse directly addresses all four gaps, including machine learning predictions, clinical thresholds, wellness metrics, and safety indicators, remains insufficiently addressed. The implementation of comprehensive systems capable of delivering real-time predictions, risk visualization, explainability, temporal analysis, and personalized recommendations within a cohesive web-based framework has not been thoroughly investigated.

MindPulse addresses all four gaps by unifying Random Forest prediction, SHAP explainability, the MHI wellness metric, the multi-signal REE, session-based temporal modelling, and Llama-3.3-70B-Versatile recommendation generation within a FastAPI–React deployment, trained and validated on 15,147 clinical records across 47 features.

## CHAPTER 3

### PROPOSED METHODOLOGY

This chapter details the complete technical methodology underlying MindPulse, covering in sequence: system architecture and data flow; dataset composition; preprocessing pipeline; ML model training, selection, and evaluation; MHI formulation; REE design; SHAP integration; temporal modelling; and LLM recommendation generation. Sufficient implementation detail is provided to support reproducibility.

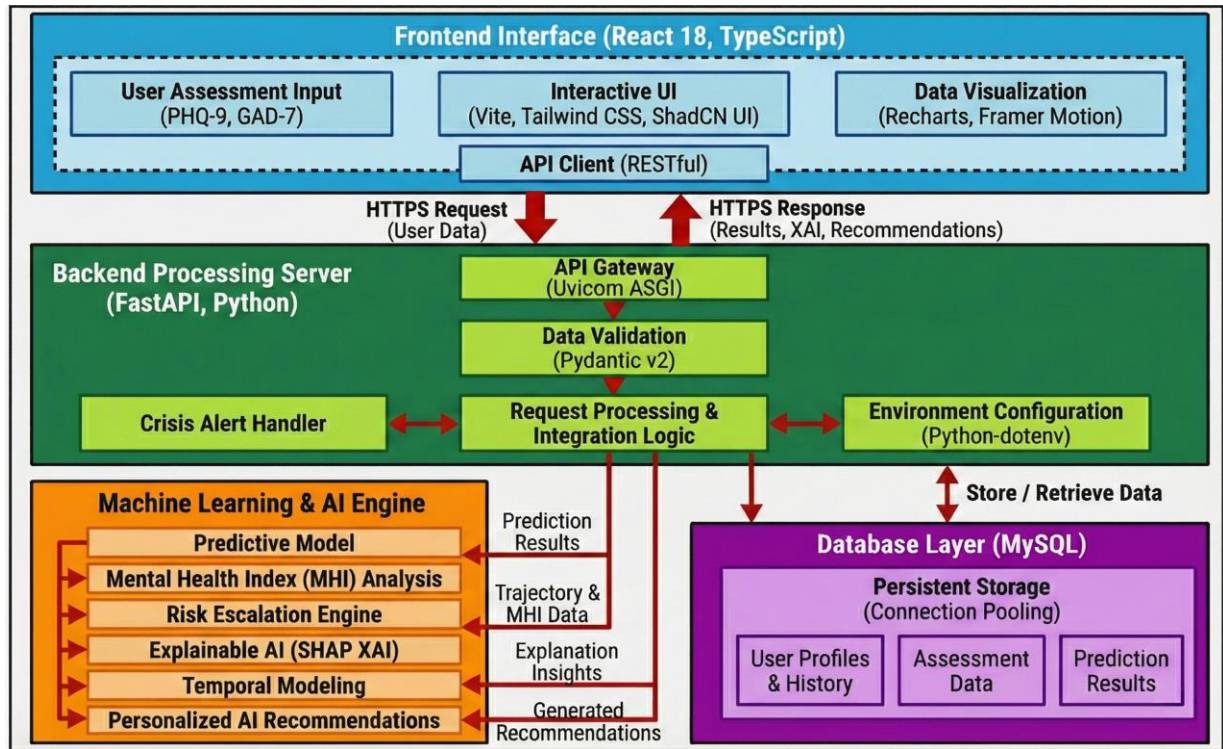
#### 3.1. Overview of Proposed System

MindPulse is a real-time AI-powered platform that detects early indicators of psychological crisis in young adults (18–30 years). Six functional components are integrated within a unified deployment framework. (1) ML Prediction: XGBoost, Random Forest, and Logistic Regression classifiers trained on standardised assessment data forecast depression and anxiety severity on a four-tier scale.

(2) Mental Health Index: PHQ-9 and GAD-7 scores are normalised and combined into a clinically tiered 0–100 wellness scale. (3) Risk Escalation Engine: Five signal streams fuse into a 0–7 escalation score with mapped clinical protocols. (4) Explainability: SHAP TreeExplainer provides per-feature attributions. (5) Temporal Modelling: Session-history analysis tracks MHI trends, deterioration, persistence, volatility, recent deterioration, risk persistence, volatility, and trajectory changes. Personalized AI Recommendations: These are LLM-generated coping strategies, wellness guidance, and crisis intervention resources, tailored to individual risk profiles.

The system comprises three tiers (Fig. 3.1). The presentation layer is a single-page application developed using React 18 and TypeScript, constructed with Vite and stylized with Tailwind

CSS and ShadCN UI, with Recharts and Framer Motion for data visualisation. The application layer is a FastAPI backend (Uvicorn ASGI, Pydantic v2) hosting all six AI components. The data layer is a MySQL database with connection pooling, storing user profiles, assessment history, and prediction records.



**Figure 3.1 System Architecture Diagram**

Frontend-backend communication occurs over HTTPS: request payloads carry PHQ-9/GAD-7 responses and auxiliary indicators; response payloads deliver risk scores, MHI values, SHAP attributions, temporal metrics, and LLM recommendations, which convey user assessment data, including PHQ-9 and GAD-7 responses, as well as additional indicators. The corresponding HTTPS responses contain prediction results and XAI recommendations. A crisis alert handler is responsible for managing the routing of critical-level alerts within the system.

### 3.2. Dataset Description

The dataset employed in this study comprised responses collected from patients utilizing standardized mental health assessment tools, specifically the PHQ-9 and GAD-7. This dataset

encompassed 15,147 records and 47 features and was sourced from the Hugging Face platform. The complete dataset details are shown in Table 3.1

**Table 3.1 Complete Dataset Information**

| Dataset               | Details                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Source                | Hugging Face                                                                                                                        |
| Dataset URL           | <a href="https://huggingface.co/datasets/KintsugiHealth/dam-dataset">https://huggingface.co/datasets/KintsugiHealth/dam-dataset</a> |
| Total Records         | 15,147                                                                                                                              |
| Total Features        | 47                                                                                                                                  |
| Primary Instruments   | PHQ-9, GAD-7                                                                                                                        |
| PHQ-9 Score Range     | 0–27                                                                                                                                |
| GAD-7 Score Range     | 0–21                                                                                                                                |
| Additional Indicators | Mood, behavior, stress, sleep, and emotional responses                                                                              |

The PHQ-9 component of the dataset assesses depressive symptoms through nine items. Each is evaluated on a scale from 0 (not at all) to 3 (almost every day), yielding a cumulative score range of 0 to 27. These nine items evaluate various aspects of depressive symptomatology, including: diminished interest or pleasure in activities, feelings of hopelessness or depression, sleep disturbances such as difficulty falling asleep, maintaining sleep, or excessive sleeping, fatigue or low energy, appetite changes such as poor appetite or overeating, negative self-perception including feelings of failure or letting others down, concentration difficulties, psychomotor changes such as noticeably slow movement or speech, or restlessness and fidgetiness, and thoughts of self-harm or suicidal ideation [13], [14].

The GAD-7 component evaluates anxiety symptoms with seven items. Each criterion is evaluated on a scale from 0 to 3, yielding a cumulative score range of 0 to 21. These items address symptoms such as nervousness, anxiety, or experiencing heightened anxiety; inability to cease or regulate concerns; disproportionate apprehension regarding several issues; challenges in achieving relaxation; restlessness or difficulties maintaining stillness; being

readily agitated or angry; and feeling apprehensive as if an ominous event may transpire [15], [16]. Alongside the PHQ-9 and GAD-7 results, the dataset encompasses further mental health indicators, including mood patterns, behavioral changes, stress levels, sleep patterns, and emotional responses.

These additional features provide comprehensive insights into individuals' psychological conditions and augment the standardized instrument scores with contextual information pertinent to mental health risk assessment. Features and their description are shown in Table 3.2.

**Table 3.2 Complete Dataset Feature Description**

| No. | Feature Name        | Data Type | Description                                                                    |
|-----|---------------------|-----------|--------------------------------------------------------------------------------|
| 1   | patient_id          | object    | Unique identifier for each patient record                                      |
| 2   | age                 | int64     | Age of the patient in years                                                    |
| 3   | ethnicity           | object    | Ethnic background of the patient                                               |
| 4   | gender              | object    | Gender of the patient                                                          |
| 5   | phq_little_interest | float64   | PHQ-9 Item 1: Little interest or pleasure in doing things (score 0–3)          |
| 6   | phq_feeling_down    | float64   | PHQ-9 Item 2: Feeling down, depressed, or hopeless (score 0–3)                 |
| 7   | phq_sleep_issues    | float64   | PHQ-9 Item 3: Trouble falling/staying asleep, or sleeping too much (score 0–3) |
| 8   | phq_low_energy      | int64     | PHQ-9 Item 4: Feeling tired or having little energy (score 0–3)                |
| 9   | phq_appetite        | int64     | PHQ-9 Item 5: Poor appetite or overeating (score 0–3)                          |
| 10  | phq_self_worth      | int64     | PHQ-9 Item 6: Feeling bad about yourself or that you are a failure (score 0–3) |
| 11  | phq_concentration   | int64     | PHQ-9 Item 7: Trouble concentrating on things (score 0–3)                      |

**Table 3.2 (continued)**

|    |                       |         |                                                                            |
|----|-----------------------|---------|----------------------------------------------------------------------------|
| 12 | phq_psychomotor       | float64 | PHQ-9 Item 8: Moving/speaking slowly or being fidgety/restless (score 0–3) |
| 13 | phq_suicidal_thoughts | int64   | PHQ-9 Item 9: Thoughts of self-harm or suicidal ideation (score 0–3)       |
| 14 | phq_total_score       | float64 | PHQ-9 total score: Sum of Items 1–9 (range 0–27)                           |
| 15 | phq_2                 | float64 | PHQ-2 score: Sum of PHQ-9 Items 1 and 2                                    |

|    |                                     |         |                                                                                         |
|----|-------------------------------------|---------|-----------------------------------------------------------------------------------------|
|    |                                     |         | (brief depression screener, range 0–6)                                                  |
| 16 | gad_nervous                         | float64 | GAD-7 Item 1: Feeling nervous, anxious, or on edge (score 0–3)                          |
| 17 | gad_uncontrollable_worry            | float64 | GAD-7 Item 2: Not being able to stop or control worrying (score 0–3)                    |
| 18 | gad_excessive_worry                 | float64 | GAD-7 Item 3: Worrying too much about different things (score 0–3)                      |
| 19 | gad_trouble_relaxing                | float64 | GAD-7 Item 4: Trouble relaxing (score 0–3)                                              |
| 20 | gad_restlessness                    | float64 | GAD-7 Item 5: Being so restless that it is hard to sit still (score 0–3)                |
| 21 | gad_irritability                    | float64 | GAD-7 Item 6: Becoming easily annoyed or irritable (score 0–3)                          |
| 22 | gad_fear                            | float64 | GAD-7 Item 7: Feeling afraid as if something awful might happen (score 0–3)             |
| 23 | gad_total_score                     | float64 | GAD-7 total score: Sum of Items 1–7 (range 0–21)                                        |
| 24 | gad_2                               | float64 | GAD-2 score: Sum of GAD-7 Items 1 and 2 (brief anxiety screener, range 0–6)             |
| 25 | duration                            | float64 | Total duration of the recorded assessment session (seconds)                             |
| 26 | prompt_vad_cut_duration             | float64 | Duration of voice activity detected in the prompt segment after cutting silence         |
| 27 | prompt_vad_trimmed_duration         | float64 | Duration of voice activity in the prompt segment after trimming                         |
| 28 | conversational_vad_cut_duration     | float64 | Duration of voice activity detected in the conversational segment after cutting silence |
| 29 | conversational_vad_trimmed_duration | float64 | Duration of voice activity in the conversational segment after trimming                 |
| 30 | mos_pred                            | float64 | Predicted Mean Opinion Score: Speech quality assessment metric                          |
| 31 | noi_pred                            | float64 | Predicted noise level in the audio recording                                            |
| 32 | loud_pred                           | float64 | Predicted loudness level of the audio recording                                         |
| 33 | dis_pred                            | float64 | Predicted speech discontinuity metric                                                   |

**Table 3.2 Continued**

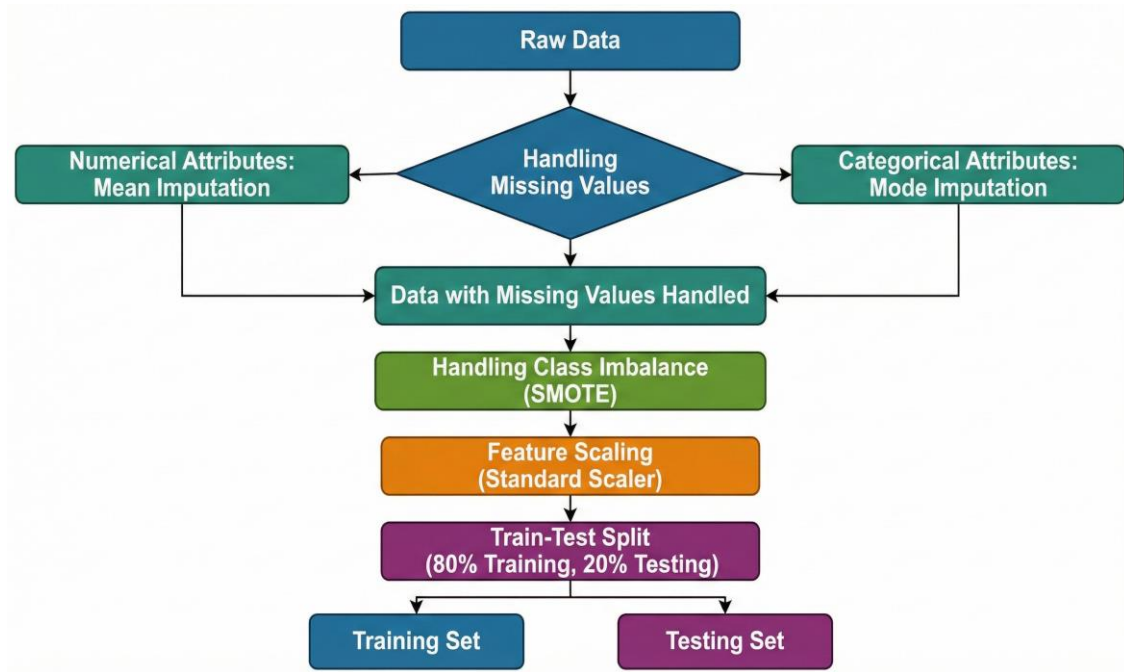
|    |                   |         |                                                                         |
|----|-------------------|---------|-------------------------------------------------------------------------|
| 34 | col_pred          | float64 | Predicted coloration metric of the audio recording                      |
| 35 | depression_binary | int64   | Binary depression indicator (0 = no depression, 1 = depression present) |
| 36 | anxiety_binary    | int64   | Binary anxiety indicator (0 = no anxiety, 1 = anxiety present)          |
| 37 | split             | object  | Dataset split assignment (e.g., train, test)                            |
| 38 | prompt            | object  | Text of the assessment prompt given to the patient                      |

|    |                     |         |                                                                                   |
|----|---------------------|---------|-----------------------------------------------------------------------------------|
|    |                     |         | (available for 9,253 records)                                                     |
| 39 | location            | object  | Geographic location of the patient (available for 9,253 records)                  |
| 40 | income_level        | object  | Self-reported income level of the patient (available for 9,253 records)           |
| 41 | english_preferred   | object  | Whether English is the patient's preferred language (available for 9,253 records) |
| 42 | preferred_lang      | object  | Patient's preferred language (available for 120 records)                          |
| 43 | anxiety_score       | float64 | Computed anxiety severity score used for model training                           |
| 44 | anxiety_severity    | int64   | Anxiety severity class label (0 = None, 1 = Mild, 2 = Moderate, 3 = Severe)       |
| 45 | depression_score    | float64 | Computed depression severity score used for model training                        |
| 46 | depression_severity | int64   | Depression severity class label (0 = None, 1 = Mild, 2 = Moderate, 3 = Severe)    |
| 47 | is_unbiased         | bool    | Flag indicating whether the record belongs to the unbiased subset of the dataset  |

Features 1-24 comprise the clinical assessment data (PHQ-9 and GAD-7, and derived scores). Features 25-34 represent audio/voice activity metrics from the assessment session. Features 35-36 are binary classification targets. Features 37-42 contain metadata and demographic information. Features 43-46 are the primary target variables for model training (anxiety and depression severity scores and class labels). Feature 47 is a data quality flag.

### 3.3. Data Preprocessing

Prior to model training, all data passed through a four-stage preprocessing pipeline (Fig. 3.2): missing-value imputation, class-imbalance correction, feature normalisation, and train-test partitioning.



**Figure 3.2 Data Preprocessing Workflow Diagram**

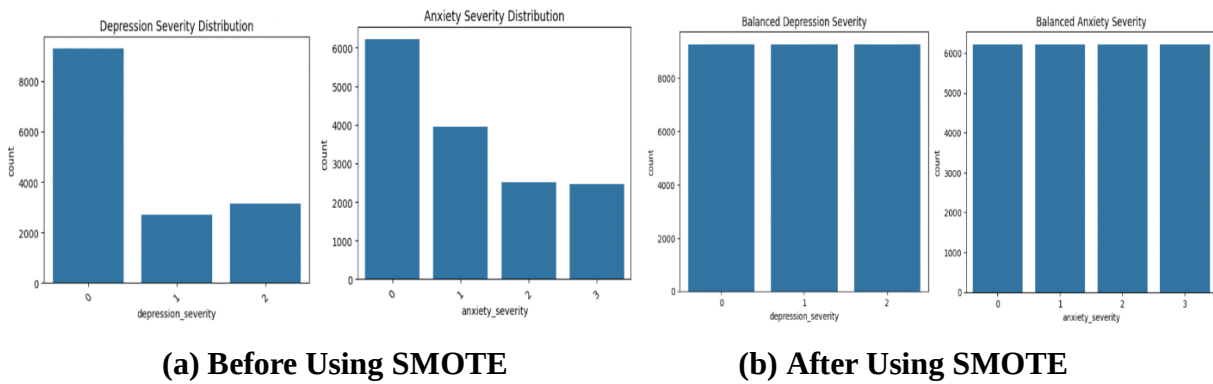
### 3.3.1. Handling Missing Values

Type-appropriate imputation addressed missing values: arithmetic mean substitution for continuous features (preserving distributional centre) and modal substitution for categorical variables (preserving predominant category). This approach maximises usable records while avoiding the information loss of listwise deletion.

### 3.3.2. Handling Class Imbalance

To address the imbalance in underrepresented risk categories, the SMOTE [17] synthesises minority-class samples through k-nearest-neighbour interpolation, producing statistically plausible observations without duplication. Applied before training, it equalises risk-category frequencies and compels classifiers to learn from all risk tiers equally, with plausible synthetic data points, thereby mitigating model bias toward majority classes and improving the model's capacity to detect high-risk mental health conditions. The utilization of SMOTE is particularly crucial in mental health datasets, where high-risk cases, those of paramount clinical interest, may be underrepresented compared to lower-risk categories. By equilibrating the class distribution before model training, SMOTE facilitates the machine learning models in learning uniformly from all risk categories, thereby enhancing sensitivity in detecting individuals at

elevated risk. The data class distribution before and after the SMOTE was applied is shown in Fig. 3.3.



**Figure 3.3 Class Distribution**

### 3.3.3. Feature Scaling

Standard Scaler normalisation (zero mean, unit variance) was applied to all continuous features, ensuring that variables on different numeric scales contribute proportionally to the optimisation objective and preventing large-range features from dominating the learning process.

### 3.3.4. Train-Test Split

An 80:20 train-test partition was applied post-SMOTE, yielding approximately 12,118 training records and 3,029 held-out test records. Splitting after balancing ensures both partitions reflect the equalised class distribution, providing an unbiased estimate of generalisation performance.

## 3.4. Machine Learning Model Development

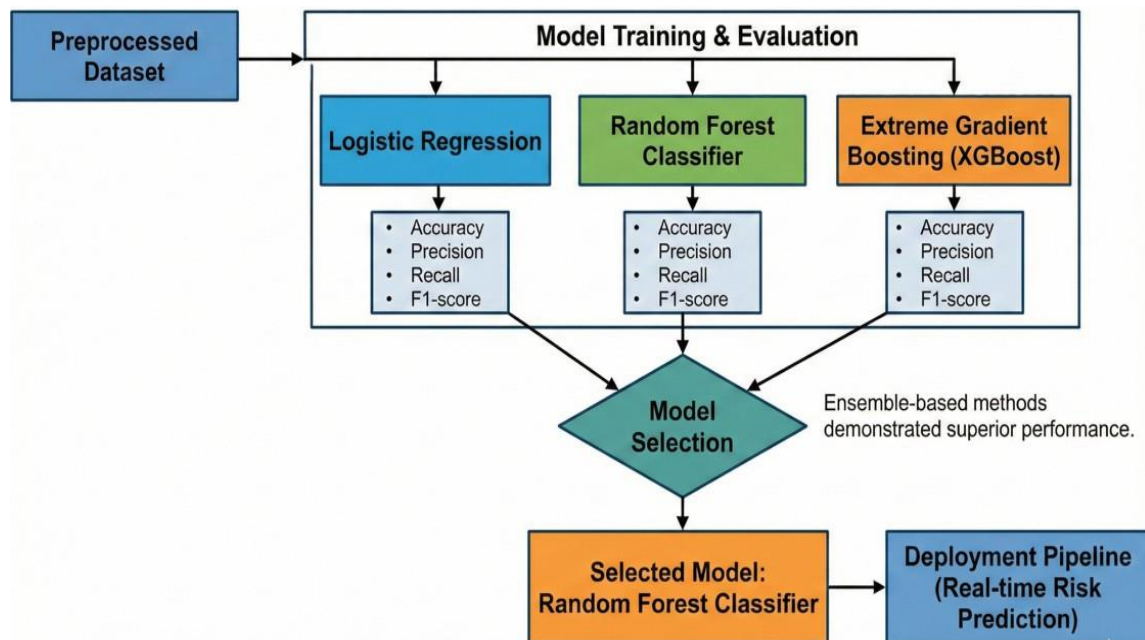
Three classifiers representing distinct modelling philosophies were benchmarked: Random Forest (bagging ensemble), Logistic Regression (linear baseline), and XGBoost (boosting ensemble). This selection enables systematic comparison across increasing model complexity while covering both interpretable and high-capacity architectures.

### 3.4.1. Logistic regression

Logistic Regression maps weighted feature combinations through a sigmoid (binary) or softmax (multinomial) function to produce class probabilities. Its primary role is to establish a transparent, interpretable performance ceiling below which ensemble methods must demonstrate clear added value.

### 3.4.2. Random Forest Classifier

The Random Forest constructs decision trees, each utilizing a bootstrapped sample with randomly selected features, subsampling at every split; majority voting determines the final class. It was selected for deployment (Fig. 3.4) based on top-tier test accuracy, overfitting resistance through ensemble averaging, mechanism, faster inference time compared to XGBoost, which requires sequential tree evaluation, ease of integration into the FastAPI-based real-time deployment architecture, and better interpretability when combined with SHAP analysis.



**Figure 3.4 Machine Learning Pipeline Diagram**

### 3.4.3. Extreme Gradient Boosting

XGBoost sequentially constructs decision trees, with each tree rectifying the prediction

residuals of the preceding ensemble. L1 and L2 regularization terms constrain model complexity, while second-order gradient optimization accelerates convergence. Although XGBoost demonstrated competitive performance in this study, the Random Forest classifier was selected for deployment due to its reduced computational overhead and ease of integration in real-time applications.

### 3.5. Model Evaluation Metrics

The evaluation of model performance was conducted utilizing four common classification metrics: accuracy, precision, recall, and F1-score. Accuracy quantifies the proportion of correctly categorized test situations across all risk categories. Although informative, it may be deceptive in the context of class imbalance. Accuracy is defined as the sum of true positives and true negatives divided by the total number of instances. While accuracy provides a thorough assessment of classification ability, it might be deceptive in situations with uneven class distributions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.1)$$

Precision (positive predictive value) quantifies the proportion of accurate positive predictions, assessing the system's ability to prevent unwarranted clinical alerts for persons not genuinely at increased risk.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.2)$$

Recall (sensitivity) measures the fraction of truly high-risk individuals the model correctly identifies. Maximising recall is clinically paramount in this context: missed detections represent the most consequential prediction failure type.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.3)$$

The F1-score represents the harmonic mean of precision and recall, yielding a singular balanced metric.

that together addresses false-positive and false-negative error rates; it is the favored summary statistic when both mistake kinds possess therapeutic importance.

$$\mathbf{F1 - Score} = 2 \times \frac{(\mathbf{Precision} \times \mathbf{Recall})}{(\mathbf{Precision} + \mathbf{Recall})} \quad (3.4)$$

Together, these four metrics provide a multifaceted classifier evaluation, ensuring both over-detection and under-detection are carefully monitored across all risk categories. Both precision and recall are clinically important.

### 3.6 Mental Health Index (MHI)

#### 3.6.1 Motivation and Clinical Grounding

Because the PHQ-9 (0–27) and GAD-7 (0–21) operate on different numeric scales, direct summation would weight them disproportionately according to range magnitude [13], [15]. The MHI resolves this by normalising both scores to [0, 1] before combining them with evidence-based differential weights, then inverting the distress index onto a 0–100 wellness continuum where higher values unambiguously indicate better psychological health.

The 0.6 depression / 0.4 anxiety weighting reflects two converging evidence streams: meta-analytic evidence that depressive symptoms exert greater functional impairment than anxiety in comorbid presentations [18], [19], and the stronger linkage between depressive severity and suicidal behaviour [20], [21]. Both streams independently support prioritising the PHQ-9 contribution.

#### 3.6.2 Motivation and Clinical Grounding

Let  $\mathbf{p}$  denote the PHQ-9 score ( $\mathbf{p} \in [0, 27]$ ) and  $\mathbf{g}$  denote the GAD-7 score ( $\mathbf{g} \in [0, 21]$ ). The MHI computation involves three sequential steps.

##### Step 1: Per Instrument Normalization

Raw scores are normalized to the unit interval  $[0, 1]$  by dividing each score by its respective maximum value:

$$p_{\text{norm}} = \frac{p}{27} \quad (3.5)$$

$$g_{\text{norm}} = \frac{g}{21} \quad (3.6)$$

This normalization ensures that both instruments contribute proportionally regardless of their different scoring ranges. After normalization, both  $p_{\text{norm}}$  and  $g_{\text{norm}}$  represent the proportion of maximum possible distress on their respective instruments.

### Step 2: Weighted Distress Composite

A weighted linear combination of the normalized scores yields the distress index  $D \in [0, 1]$ :

$$D = 0.6 p_{\text{norm}} + 0.4 g_{\text{norm}} \quad (3.7)$$

The weights  $w_1 = 0.6$  and  $w_2 = 0.4$  reflect the differential impact of depression on functional impairment and satisfy the constraint:

$$w_1 + w_2 = 1.0 \quad (3.8)$$

The distress index  $D$  represents the weighted proportion of the maximum possible combined distress, where  $D = 0$  indicates no distress on either instrument and  $D = 1$  indicates maximum distress on both instruments.

### Step 3: Inversion to Wellness Scale

The distress index is inverted and scaled to the **[0, 100]** range to produce the wellness-oriented MHI:

$$MHI = (1 - D) \times 100 \quad (3.9)$$

The result is clamped to ensure it remains within the valid range:

$$MHI = \max(0, \min(100, MHI)) \quad (3.10)$$

The inversion ensures that higher scores on the Mental Health Inventory (MHI) correspond to enhanced mental health, characterized by diminished distress, thereby aligning with the intuitive understanding of a "wellness" scale. An MHI score of 100 indicates the absence of reported symptoms on either instrument, whereas an MHI score of 0 signifies maximum severity on both.

### 3.6.3 Motivation and Clinical Grounding

The continuous MHI score is mapped to four discrete wellness tiers using clinical severity thresholds, as shown in Table 3.2.

**Table 3.3 MHI Wellness Tiers and Clinical Interpretation**

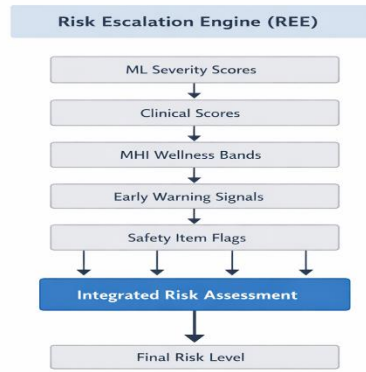
| MHI Range | Tier     | D Range   | Clinical Interpretation                               |
|-----------|----------|-----------|-------------------------------------------------------|
| 75–100    | Good     | 0.00–0.25 | Minimal-to-mild symptoms; no clinical concern         |
| 50–74     | Moderate | 0.25–0.50 | Moderate distress; monitoring recommended             |
| 25–49     | Poor     | 0.50–0.75 | Significant distress; professional evaluation advised |
| 0–24      | Critical | 0.75–1.00 | Severe distress; immediate clinical intervention      |

## 3.7 Risk Escalation Engine (REE)

### 3.7.1 Design Rationale

The Risk Escalation Engine (REE) constitutes the central clinical risk assessment module of the

MindPulse system. It addresses the limitation of relying exclusively on machine learning (ML) predictions for risk determination by integrating multiple complementary signal sources. The REE synthesizes five components: ML severity scores, raw clinical scores from standardized instruments, Mental Health Index (MHI) wellness band contributions, early warning signals, and safety item flags. The complete diagram is depicted in Fig. 3.5. This multi-component approach ensures that the final risk assessment reflects both the ML model's learned patterns and established clinical thresholds.



**Figure 3.5 Risk Escalation Engine Block Diagram**

### 3.7.2 Component 1: ML Severity Score ( $S_{ML}$ )

The Random Forest classifier generates severity predictions for depression  $\mathbf{dep}_{pred} \in \{0, 1, 2, 3\}$  and anxiety  $\mathbf{anx}_{pred} \in \{0, 1, 2, 3\}$ , corresponding to none, mild, moderate, and severe categories. These discrete predictions are converted to continuous risk scores through asymmetric mapping functions that emphasize higher severity levels:

$$S_{ML} = f_{dep}(\mathbf{dep}_{pred}) + f_{anx}(\mathbf{anx}_{pred}) \quad (3.11)$$

The higher ceiling for depression (4.0 vs. 2.5 for anxiety) reflects the stronger association of depression with functional impairment and suicide risk [20], [21]. The asymmetric mapping ensures that severe levels contribute disproportionately more to the risk score than mild levels, reflecting the nonlinear relationship between symptom severity and clinical risk. The maximum value of  $S_{ML}$  is 6.5 (4.0 + 2.5) as shown in Table 3.3, occurring when both depression and anxiety are predicted as severe.

**Table 3.4 ML Severity-to-Risk Conversion Functions**

| Level | Label    | $f_{\text{dep}}$ | $f_{\text{anx}}$ |
|-------|----------|------------------|------------------|
| 0     | None     | 0                | 0                |
| 1     | Mild     | 1.0              | 0.5              |
| 2     | Moderate | 2.5              | 1.5              |
| 3     | Severe   | 4.0              | 2.5              |

**Component 2: Raw Clinical Score ( $S_{\text{raw}}$ )**

The raw clinical score directly employs the total scores from the PHQ-9 and GAD-7, aligning them with risk contributions according to DSM-consistent severity thresholds [14], [15]. This feature ensures that the REE integrates clinical severity data from the raw instrument scores, independent of the predictions generated by the ML model.

The PHQ-9 and GAD-7 totals map to risk contributions using DSM-aligned thresholds.

$$S_{\text{phq}} = \begin{cases} 2.0, & p \geq 20 \\ 1.5, & p \geq 15 \\ 1.0, & p \geq 10 \\ 0.5, & p \geq 5 \\ 0, & \text{otherwise} \end{cases} \quad (3.12)$$

$$S_{\text{gad}} = \begin{cases} 1.5, & g \geq 15 \\ 1.0, & g \geq 10 \\ 0.5, & g \geq 5 \\ 0, & \text{otherwise} \end{cases} \quad (3.13)$$

$$S_{\text{raw}} = S_{\text{phq}} + S_{\text{gad}} \quad (3.14)$$

The maximum value of  $S_{\text{raw}}$  was 3.5, with PHQ-9 having a higher ceiling than GAD-7 due to

functional burden association.

### Component 3: MHI Wellness Band ( $S_{\text{MHI}}$ )

The MHI wellness tier contributes a score  $S_{\text{MHI}}$  to the REE based on the wellness tier classification.

$$S_{\text{MHI}} = \begin{cases} 2.0, & \text{tier} = \text{Critical} \\ 1.0, & \text{tier} = \text{Poor} \\ 0.5, & \text{tier} = \text{Moderate} \\ 0, & \text{tier} = \text{Good} \end{cases} \quad (3.15)$$

### Component 4: Early Warning Signal

$$S_{\text{warn}} = \begin{cases} 3.0, & \text{level} = \text{Critical} \\ 1.5, & \text{level} = \text{Warning} \\ 0.5, & \text{level} = \text{Watch} \\ 0, & \text{level} = \text{None} \end{cases} \quad (3.16)$$

### Component 5: Safety Item Flags ( $S_{\text{item}}$ )

$$E_{\text{raw}} = 0.35 S_{\text{ML}} + 0.25 S_{\text{raw}} + 0.10 S_{\text{MHI}} + 0.20 S_{\text{warn}} + 0.10 S_{\text{item}} \quad (3.17)$$

$\text{PHQ}_9 \geq 2$  indicates a critical safety signal requiring substantial risk contribution.

The components are combined using fixed weights based on diagnostic validity.

With  $E_{\text{max}} \approx 6.65$ , the score normalizes to  $[0,7]$ :

$$E = \min\left(E_{\text{raw}} \times \frac{7.0}{6.65}, 7.0\right) \quad (3.18)$$

The escalation score  $E$  is mapped to five risk levels with corresponding clinical action protocols, as shown in Table 3.4.

**Table 3.5 Risk Escalation Thresholds and Clinical Protocols**

| Score Range        | Level    | Action   | Protocol                                    |
|--------------------|----------|----------|---------------------------------------------|
| $E \geq 5.5$       | Critical | ESCALATE | Immediate referral; crisis support          |
| $3.8 \leq E < 5.5$ | High     | URGENT   | Consult within 48–72 hours                  |
| $2.4 \leq E < 3.8$ | Moderate | MONITOR  | Counseling within 1–2 weeks                 |
| $1.2 \leq E < 2.4$ | Low-Mod  | PREVENT  | Lifestyle interventions + re-screen 4 weeks |
| $E < 1.2$          | Low      | WELLNESS | Maintain wellness; re-screen after 3 months |

### 3.8 SHAP-Explainable AI Integration

MindPulse integrates SHAP TreeExplainer to make every risk prediction auditable [6], [7]. Grounded in Shapley values from cooperative game theory, the framework uniquely and axiomatically attributes each feature's contribution to the deviation of a specific prediction from the population baseline.

$$f(x) = \phi_0 + \sum_{i=1}^n \phi_i x_i \quad (3.19)$$

where  $\phi_0$  represents the base value and  $\phi_i$  denotes the SHAP value for feature  $i$ . Positive SHAP values signify that a trait elevates the forecast of severity (augmenting projected risk), whereas negative values denote a protective influence (diminishing predicted risk).

The SHAP module delivers three output types: (a) a global importance summary ranking features by mean |SHAP value| for model-level validation; (b) per-user waterfall charts showing feature-specific contributions for each individual assessment; and (c) a ranked list of the primary symptom drivers of that user's risk score, providing an actionable focus for clinical discussion and self-management.

### 3.9 Temporal Modeling for Longitudinal Trend Analysis

Cross-sectional assessments capture only a momentary state; they cannot detect trajectories of

gradual deterioration that frequently precede crises. The temporal modelling module analyses l

ongitudinal dynamics across  $N$  assessment sessions ( $N \geq 2$ ) to provide longitudinal insights. Each session is represented as a tuple:

$$h_t = (p_t, g_t, M_t, r_t) \tag{3.20}$$

where  $p_t$ ,  $g_t$ ,  $M_t$ , and  $r_t$  denote the PHQ-9 score, GAD-7 score, MHI, and risk level at session  $t$ , respectively [3], [12].

### 3.9.1 Global Trend Computation

The global trajectory is computed by comparing the MHI at the first and last sessions:

$$\Delta MHI = M_N - M_1 \tag{3.21}$$

The trend is classified in accordance with the subsequent thresholds as illustrated in Table 3.5.

**Table 3.6 Trend Classification Thresholds**

| Condition         | Classification | Interpretation              |
|-------------------|----------------|-----------------------------|
| $\Delta MHI > 8$  | Improving      | Overall wellness increasing |
| $\Delta MHI < -8$ | Deteriorating  | Overall wellness decreasing |
| Otherwise         | Stable         | No significant change       |

### 3.9.2 Recent Deterioration Detection

Recent deterioration is detected using a sliding window comparison that compares recent sessions to earlier sessions:

$$n = \min(3, \lfloor N/2 \rfloor) \tag{3.22}$$

$$\Delta MHI_r = \frac{1}{n} \sum_{t=N-n+1}^N M_t - \frac{1}{n} \sum_{t=1}^n M_t \quad (3.23)$$

A deterioration flag  $D_{\text{flag}}$  is raised when:

$$D_{\text{flag}} = (\Delta MHI_r < -10) \vee [(\Delta PHQ > 8) \wedge (\Delta GAD > 5)] \quad (3.24)$$

### 3.9.3 Risk Persistence

Risk persistence measures the length of the longest consecutive streak of the same risk level at the end of the session history:

$$\text{streak} = \max\{k: r_{N-k+1} = \dots = r_N\} \quad (3.25)$$

### 3.9.4 Wellness Volatility

Volatility quantifies the variability of MHI scores across sessions using the standard deviation:

$$\sigma_M = \sqrt{\frac{1}{N} \sum_{t=1}^N (M_t - \bar{M})^2} \quad (3.26)$$

The wellness volatility is classified based on the threshold, as shown in Table 3.5.

**Table 3.7 Volatility Classification Thresholds**

| Condition              | Classification      |
|------------------------|---------------------|
| $\sigma_M > 15$        | High volatility     |
| $7 < \sigma_M \leq 15$ | Moderate volatility |

|                   |                |
|-------------------|----------------|
| $\sigma_M \leq 7$ | Low volatility |
|-------------------|----------------|

### 3.9.5 Risk Trajectory

Risk trajectory analysis tracks whether the overall risk level has worsened by comparing the rank of the final risk level with the initial risk level:

$$\text{risk}_{\text{worsened}} = (\text{rank}(r_N) > \text{rank}(r_1)) \quad (3.27)$$

Additional metrics computed by the temporal module include peak, minimum, and mean MHI values, as well as PHQ-9 and GAD-7 deltas to detect deterioration patterns.

### 3.10 Personalized AI Recommendations

To translate risk outputs into actionable support, MindPulse passes each user's risk tier, MHI score, and top SHAP-ranked symptom contributors as a structured prompt to Llama-3.3-70B-Versatile [8]–[11]. The model generates four output categories: (a) coping strategies targeting the identified symptom drivers; (b) wellness maintenance practices calibrated to the current MHI tier; (c) preventive behaviours appropriate to the risk level; and (d) crisis resources and immediate support contacts for Critical or High escalation cases.

Prompt engineering is deliberate: risk level, MHI score with tier label, and the top SHAP contributors with attribution directions are serialised into a structured context block before the instruction. This constrained approach steers the LLM toward clinically grounded, profile-specific responses, limiting the risk of generic or hallucinatory outputs by anchoring the model to verifiable numerical clinical facts.

# CHAPTER 4

## RESULTS

This chapter presents and interprets the experimental outcomes for each of MindPulse's six functional components. The implementation environment is summarised in Table 4.1, followed by results for: ML benchmarking and model selection; MHI analysis; REE evaluation; SHAP explainability; temporal trajectory monitoring; and LLM-driven personalised recommendations.

Table 4.1 Implementation Environment

| Component            | Specification                                                         |
|----------------------|-----------------------------------------------------------------------|
| Training Platform    | Google Colaboratory (NVIDIA Tesla T4 GPU)                             |
| Development Machine  | Alienware m15 Ryzen Ed. R5 (AMD Ryzen 7 5800H, 16 GB RAM)             |
| Operating System     | Windows 11 Home 64-Bit                                                |
| IDE                  | Visual Studio Code                                                    |
| Programming language | Python 3.14.3                                                         |
| ML Libraries         | scikit-learn 1.8.0, SHAP 0.50.0, NumPy 2.4.2, Pandas 3.0.1            |
| Backend              | FastAPI 0.129.0, Uvicorn 0.41.0, Pydantic 2.12.5                      |
| Frontend             | React 18, TypeScript, Vite, Tailwind CSS, ShadCN UI (Node.js v25.6.1) |
| LLM Integration      | Llama-3.3-70B-Versatile via Groq API (SDK 1.0.0)                      |
| Database             | MySQL (mysql-connector-python 9.6.0)                                  |

4.1 Experimental Setup

All experiments used the 15,147-record, 47-feature Hugging Face dataset. After mean/mode imputation, SMOTE balancing, and Standard Scaler normalisation, data were partitioned 80/20, yielding approximately 12,118 training and 3,029 test records.

All three classifiers were trained on Google Colaboratory (NVIDIA Tesla T4) using scikit-learn 1.8.0 under Python 3.14.3. Random Forest was selected for deployment based on accuracy, inference efficiency, and TreeExplainer compatibility. SHAP v0.50.0 generated feature attribution analyses on the deployed model.

The temporal module required  $N \geq 2$  sessions. Llama-3.3-70B-Versatile was accessed via Groq SDK v1.0.0. The full stack (FastAPI v0.129.0, Uvicorn v0.41.0, React 18 + TypeScript + Vite, and MySQL) was developed in VS Code. Full configuration details appear in Table 4.2.

Table 4.2 Experimental Configuration Summary

| Parameter              | Value                                                         |
|------------------------|---------------------------------------------------------------|
| Dataset Size           | 15,147 records, 47 features                                   |
| Train-Test Split       | 80% training (~12,118 records) / 20% testing (~3,029 records) |
| Missing Value Handling | Mean imputation (numerical), Mode imputation (categorical)    |
| Class Balancing        | SMOTE (Synthetic Minority Over-sampling Technique)            |
| Feature Scaling        | Standard Scaler (zero mean, unit variance)                    |
| Models Evaluated       | Logistic Regression, Random Forest, XGBoost                   |
| Selected Model         | Random Forest Classifier                                      |
| Explainability         | SHAP (TreeExplainer for Random Forest)                        |
| Temporal Modeling      | Session-based ( $N \geq 2$ sessions)                          |
| LLM Integration        | Llama-3.3-70B-Versatile via Groq API                          |

|                      |                                       |
|----------------------|---------------------------------------|
| Backend API          | FastAPI with Uvicorn ASGI server      |
| Frontend             | React 18 + TypeScript + Vite          |
| Database             | MySQL with connection pooling         |
| Training Environment | Google Colab with NVIDIA Tesla T4 GPU |

## 4.2 Performance Evaluation Metrics

The model's performance was evaluated using accuracy, F1-score, recall, and precision. Recall carries particular clinical weight because undetected high-risk cases represent the most consequential error, while precision matters to avoid generating disproportionate clinical burden for low-risk individuals.

## 4.3 Model Performance Comparison

Table 4.3 compares all three classifiers on the reserved test set. All models achieved strong performance.

**Table 4.3 Comparative Analysis of Machine Learning Model Performance**

| Model               | Accuracy | Precision | Recall | F1-score |
|---------------------|----------|-----------|--------|----------|
| XGBoost             | 1.00     | 1.00      | 1.00   | 1.00     |
| Random Forest       | 1.00     | 1.00      | 1.00   | 1.00     |
| Logistic Regression | 0.99     | 0.99      | 1.00   | 1.00     |

XGBoost and Random Forest both attained perfect or near-perfect scores across all four metrics, correctly classifying virtually every test instance. Logistic Regression achieved marginally lower accuracy (0.99) while maintaining perfect recall and F1-score, confirming that the preprocessed feature space carries sufficient discriminative signal even for a linear classifier.

These results align with the literature confirming ensemble superiority on mental health classification tasks [5]. The parity between XGBoost and Random Forest indicates that the dataset's feature structure accommodates both bagging and boosting approaches effectively.

Table 4.4 provides training and testing accuracy figures, confirming generalisation without overfitting across all classifiers.

**Table 4.4 Training and Testing Accuracies of Machine Learning Models**

| Model               | Training Accuracy | Testing Accuracy |
|---------------------|-------------------|------------------|
| Logistic Regression | 0.99              | 0.99             |
| Random Forest       | 1.00              | 1.00             |
| XGBoost             | 1.00              | 1.00             |

Ensemble models achieved 1.00 on both training and test partitions; Logistic Regression attained 0.99 on both. The training-testing consistency rules out overfitting. The near-perfect ensemble figures most plausibly reflect strong class separability in the preprocessed feature space, a consequence of the clean SMOTE-post-split pipeline, though external validation on independent datasets remains an important future validation step.

#### 4.3.1 Model Selection Rationale

Despite equivalent accuracy, Random Forest was selected over XGBoost for three deployment-specific reasons: parallel tree evaluation yields lower inference latency than XGBoost's sequential dependency structure; FastAPI integration is straightforward with joblib serialisation; and SHAP TreeExplainer computes exact (not approximate) Shapley values for Random Forest, guaranteeing attribution fidelity in the explainability layer.

#### 4.4 Mental Health Index Analysis

The MHI maps each user's combined PHQ-9/GAD-7 profile onto a 0–100 wellness scale partitioned into four tiers: Good (75–100), Moderate (50–74), Poor (25–49), and Critical (0–24). An SVG arc gauge (Fig. 4.1) in the React dashboard renders the score with colour-coded band segments and a rotating indicator, giving users an immediately intuitive read of their psychological status.

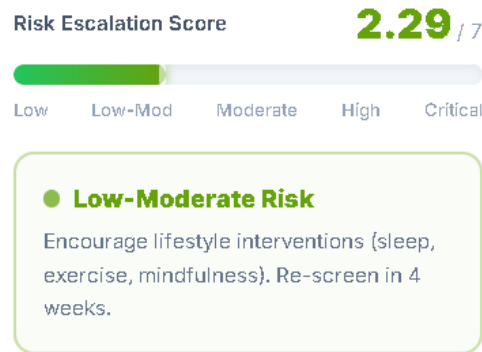


**Figure 4.1 MHI Score Gauge Visualization**

MHI values and tier labels are persisted in the MySQL session store, feeding the temporal module's longitudinal calculations. The MHI serves a dual function: an accessible wellness summary for users, and a structured numeric time series for automated trajectory analysis across sessions.

#### 4.5 Risk Escalation Engine Evaluation

The REE fuses five signal streams into a normalised escalation score  $E \in [0, 7]$ . Evaluation confirmed the system's ability to stratify users across five actionable risk tiers (Fig. 4.2), each linked to a distinct clinical intervention protocol.

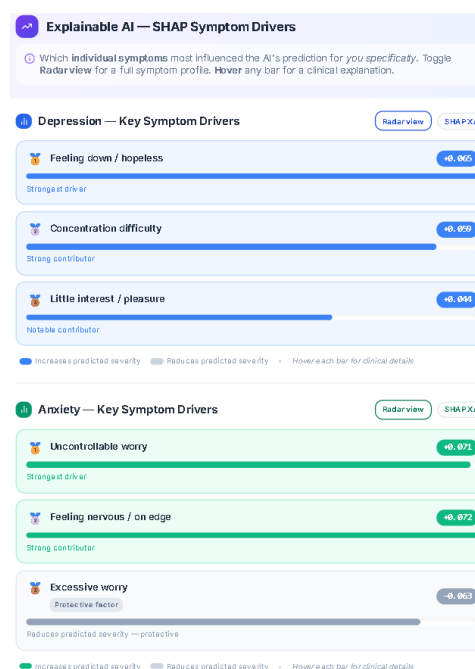


**Figure 4.2 Risk Escalation Score**

Multi-signal weighting ensures that both learned ML patterns and validated clinical thresholds jointly govern risk determination. The standalone PHQ-9 Item 9 suicidal ideation flag can elevate any assessment to Critical regardless of the ML severity score, reflecting the clinical non-negotiability of safety signals over statistical inference.

## 4.6 Explainable AI Outcomes

SHAP analysis on the deployed Random Forest model identified the symptom-level drivers of both depression and anxiety severity predictions (Fig. 4.3).



**Figure 4.3 SHAP XAI Results**

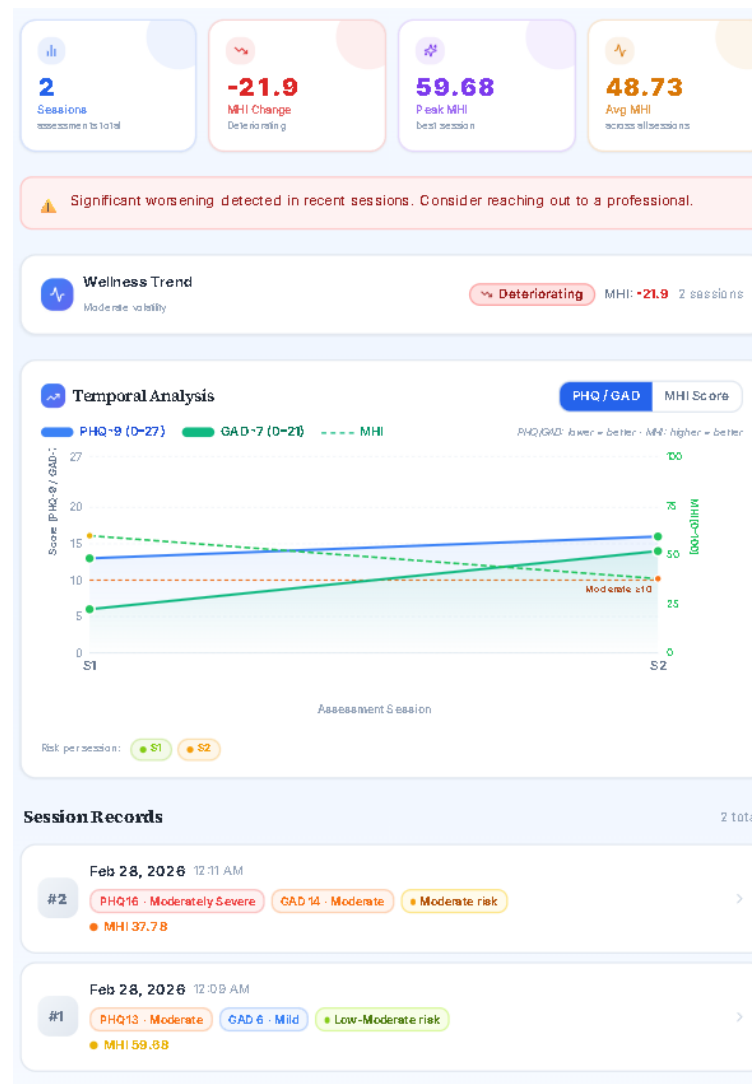
For depression severity, the three highest-SHAP features were: persistent low mood/hopelessness (PHQ Item 2), concentration difficulties (PHQ Item 7), and anhedonia (PHQ Item 1), all of which are core DSM criteria for major depressive disorder [13], [14], confirming that the model exploits clinically coherent symptom signals.

For anxiety severity, uncontrollable worry (GAD Item 2) was the strongest driver, followed by pervasive nervousness (GAD Item 1) and excessive worry (GAD Item 3), all cardinal criteria for generalised anxiety disorder [15], [16], further confirming the model's clinical validity.

Per-user SHAP outputs serve three practical functions: they inform individuals which symptoms are driving their risk score, supporting self-management; they enable clinicians to verify clinical coherence before acting on assessments; and they provide a structured basis for patient-clinician conversations about symptom priorities.

#### **4.7 Temporal Analysis for Longitudinal Trend Monitoring**

The temporal modelling module was evaluated using session-sequence data to analyse longitudinal health trajectories [3], [12]. MHI changes were tracked across sessions and categorised as Improving, Stable, or Deteriorating based on the  $\Delta MHI$  threshold values as shown in Fig. 4.4.



**Figure 4.4 Temporal Analysis for Longitudinal Trend Monitoring**

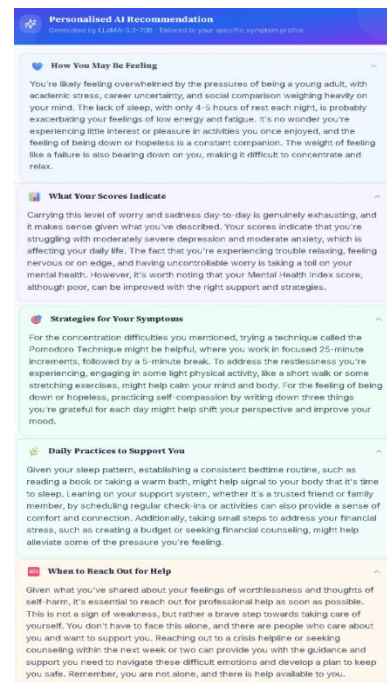
The sliding-window comparator reliably flagged recent deterioration by contrasting mean MHI from recent vs. earlier sessions. Risk persistence streaks facilitated the identification of individuals experiencing prolonged high-risk states. The measurement of wellness volatility served as an indicator of the stability of mental health status over time, with high volatility potentially signifying unstable conditions that necessitate closer monitoring. The analysis of risk trajectory monitored whether overall risk levels deteriorated throughout the assessment period.

Together, these five temporal metrics provide a richer clinical picture than any single-session snapshot: they surface deterioration trajectories, persistent risk burdens, and unstable wellness

patterns that cross-sectional screening systematically misses, enabling proactive intervention before conditions reach crisis severity.

## 4.8 Personalized AI Recommendations

Llama-3.3-70B-Versatile generated structured, four-category personalised outputs (Fig. 4.5) for each user [8], [9]: targeted coping strategies addressing their highest-SHAP symptoms; behavioural wellness recommendations calibrated to their MHI tier; risk-level-appropriate preventive guidance; and for Critical or High escalation, crisis support contacts and safety planning resources.



**Figure 4.5 Personalized AI Recommendations**

Together, MindPulse's six integrated components, namely ML prediction, SHAP explainability, multi-signal REE, MHI wellness quantification, temporal trajectory analysis, and LLM recommendation generation, cover the complete assessment-to-action cycle within a single production-deployed digital health platform.

## CHAPTER 5

### CONCLUSION

MindPulse demonstrates that early detection of mental health crises in young adults can be achieved through principled integration of multiple AI modalities within a production-deployed digital platform. By combining ML risk classification, SHAP interpretability, the MHI wellness metric, the multi-signal REE, longitudinal temporal modelling, and LLM-personalised guidance, the system addresses the core limitations of existing solutions: opacity, temporal blindness, and the gap between risk detection and actionable support.

The experimental results validate each system component. Random Forest achieved near-perfect accuracy across training and test partitions. SHAP attributions identified clinically coherent symptom drivers aligned with DSM criteria, providing the auditable reasoning providers require when acting on AI-generated risk assessments. The model's predictions identify key symptoms that influence risk classifications and enable clinicians and users to comprehend the rationale underlying each assessment. The MHI proved effective as a unified wellness summary, resolving PHQ-9/GAD-7 scale incompatibility through principled normalisation and clinically grounded differential weighting. The resulting 0–100 metric enables intuitive longitudinal tracking and communicates health status accessibility to non-specialist users.

The Risk Escalation Engine successfully synthesised five complementary signal types into a balanced composite score. The five-tier stratification scheme, with each tier mapped to a specific clinical action protocol, enables immediately operationalizable decisions, including non-negotiable escalation for PHQ-9 Item 9 suicidal ideation, with prioritization. The five-tier risk stratification, accompanied by corresponding intervention protocols, provides actionable guidance for both automated systems and clinical decision-making.

The temporal module demonstrated that session-sequence analysis surfaces clinically actionable patterns invisible to single-point assessment: deterioration trajectories, sustained high-risk states, and volatile wellness fluctuations that cross-sectional screening misses, enabling proactive intervention before conditions escalate to crisis.

LLM integration successfully bridged the detection-to-support gap: structured prompts anchored to individual SHAP profiles and risk tiers yielded specific, relevant guidance across all four output categories, representing a qualitative improvement over the generic, rule-based recommendations common in existing digital mental health tools.

The FastAPI–React deployment validates MindPulse's operational viability beyond prototype status: real-time inference, interactive dashboards, persistent session storage, and automated crisis alerts function within a single deployable stack suitable for university, corporate, and clinical healthcare environments.

MindPulse represents a meaningful step toward accessible, interpretable, and comprehensive AI-assisted mental health monitoring. Its contribution lies not in any single component but in their deliberate integration: each module, from ML classifier to LLM, serves a specific clinical purpose, and together they constitute a coherent, end-to-end system moving from raw questionnaire responses to personalised, actionable mental health support.

## CHAPTER 6

### FUTURE SCOPE

This chapter outlines five concrete directions for extending MindPulse, each addressing a specific limitation of the present implementation.

MindPulse was trained and evaluated on a single English-language Hugging Face dataset. Future work should incorporate data spanning diverse geographic regions, cultural backgrounds, and clinical settings to broaden demographic coverage. Cross-dataset validation would provide the external validity evidence required for broader clinical adoption.

The integration of additional sensing modalities represents a major enhancement opportunity. Physiological wearable signals (heart rate variability, galvanic skin response, sleep architecture) and passively collected digital behavioural data (social media engagement patterns, activity levels) would complement self-reported questionnaire scores with objective, continuously sampled biological and behavioural indicators, advancing MindPulse toward a genuinely multimodal architecture. The incorporation of these modalities would advance the system toward a genuinely multimodal architecture, leveraging diverse data sources for a more comprehensive risk assessment.

The temporal module currently classifies past trends reactively. Future work should implement prospective forecasting (e.g., LSTM or Prophet models) to project MHI trajectories and predict future deterioration windows, alongside change-point detection to identify the exact session at which a clinically significant health transition occurred.

LLM recommendations are currently conditioned on snapshot inputs, not response history. A reinforcement learning layer could learn from each user's engagement and outcomes to adaptively optimise recommendation content, timing, and delivery intensity, moving the system from personalised-but-fixed guidance toward a dynamic adaptive intervention engine.

Wider clinical use necessitates HL7 FHIR-compliant connection with Electronic Health Record systems, prospective randomized clinical validation studies, and clinician-oriented dashboards designed to enhance documentation workflows. Bidirectional feedback mechanisms, which would allow clinicians to annotate, correct, or endorse system outputs, would enable continuous supervised improvement of both ML models and LLM prompt strategies over time.

## CHAPTER 7

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## APPENDIX A

### CODING

This section contains details on the software and tools used in the project.

#### ML model Building Code Sample:

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay, classification_report

log_dep = LogisticRegression(max_iter=500)
log_dep.fit(X_train, y_dep_train)

pred_log_dep = log_dep.predict(X_test)

log_anx = LogisticRegression(max_iter=500)
log_anx.fit(X_train2, y_anx_train)

pred_log_anx = log_anx.predict(X_test2)

rf_dep = RandomForestClassifier(n_estimators=200, random_state=42)
rf_dep.fit(X_train, y_dep_train)

pred_rf_dep = rf_dep.predict(X_test)

rf_anx = RandomForestClassifier(n_estimators=200, random_state=42)
rf_anx.fit(X_train2, y_anx_train)

pred_rf_anx = rf_anx.predict(X_test2)

xgb_dep = XGBClassifier(use_label_encoder=False, eval_metric='mlogloss')
xgb_dep.fit(X_train, y_dep_train)

pred_xgb_dep = xgb_dep.predict(X_test)

xgb_dep = XGBClassifier(use_label_encoder=False, eval_metric='mlogloss')
xgb_dep.fit(X_train, y_dep_train)

pred_xgb_dep = xgb_dep.predict(X_test)

import joblib

joblib.dump(rf_dep, "depression_model.pkl")

['depression_model.pkl']

joblib.dump(rf_anx, "anxiety_model.pkl")

['anxiety_model.pkl']
```

## Web App Code Sample:

```
# =====
# /predict ENDPOINT
# =====
@app.post("/predict")
def predict(data: MentalInput):

    values = build_feature_vector(data)

    dep_pred = int(dep_model.predict(values)[0])
    anx_pred = int(anx_model.predict(values)[0])

    # — Clinical override: cap ML predictions to clinically valid ranges —
    # PHQ-9: 0-4→Minimal(0), 5-9→Mild(1), 10-14→Moderate(2), 15+→Severe(3)
    # GAD-7: 0-4→Minimal(0), 5-9→Mild(1), 10-14→Moderate(2), 15+→Severe(3)
    def _phq_clinical_max(score: int) -> int:
        if score <= 4: return 0
        if score <= 9: return 1
        if score <= 14: return 2
        return 3

    def _gad_clinical_max(score: int) -> int:
        if score <= 4: return 0
        if score <= 9: return 1
        if score <= 14: return 2
        return 3

    dep_pred = min(dep_pred, _phq_clinical_max(data.phq_total_score))
    anx_pred = min(anx_pred, _gad_clinical_max(data.gad_total_score))

    dep_probs = dep_model.predict_proba(values)[0]
    anx_probs = anx_model.predict_proba(values)[0]
    confidence = round(((float(max(dep_probs)) + float(max(anx_probs))) / 2) * 100, 2)

    # Domain-filtered SHAP: depression model → only PHQ symptom features
    #                               anxiety model → only GAD symptom features
    top_dep_features = get_shap_top_features(dep_explainer, values, n=5, domain_filter=PHQ_FEATURES)
    top_anx_features = get_shap_top_features(anx_explainer, values, n=5, domain_filter=GAD_FEATURES)

    # All SHAP values for full chart visualization (no domain filter)
    all_dep_shap = get_all_shap_values(dep_explainer, values)
    all_anx_shap = get_all_shap_values(anx_explainer, values)

    mhi = compute_mhi(data.phq_total_score, data.gad_total_score)
    warning = early_warning_detection(data)
    escalation = risk_escalation_engine(
        dep_pred, anx_pred, mhi, warning,
        data.phq_total_score, data.gad_total_score,
        data=data,
    )

    ai_text = generate_ai_recommendation(
        dep_pred=dep_pred,
        anx_pred=anx_pred,
        phq_score=data.phq_total_score,
        gad_score=data.gad_total_score,
        mhi=mhi,
        warning=warning,
        escalation=escalation,
        top_dep_features=top_dep_features,
        top_anx_features=top_anx_features,
        data=data,
    )

    # — Persist to DB (all fields, AI text EXCLUDED from storage) —
    try:
        conn = get_db_connection()
        cur = conn.cursor()
        cur.execute("""
            INSERT INTO assessments (
                user_name, user_email, age, gender,
                education, occupation, sleep_pattern, stress_source,
                mental_history, support_system, stressful_events,
                phq_q1, phq_q2, phq_q3, phq_q4, phq_q5,
                phq_q6, phq_q7, phq_q8, phq_q9, phq_score,
                gad_q1, gad_q2, gad_q3, gad_q4, gad_q5, gad_q6, gad_q7,
                gad_score,
                depression_level, anxiety_level, confidence_pct, risk_level,
                mhi_score, mhi_label, escalation_score, warning_level,
                created_at
            ) VALUES (
                %s, %s, %s, %s,
                %s, %s, %s, %s,
                %s, %s, %s,
                %s, %s, %s, %s, %s,
                %s, %s, %s, %s, %s,
                %s, %s, %s, %s,
                %s, %s, %s, %s,
                %s, %s, %s, %s,
                %s, %s, %s, %s,
                %s
            )
        """)
    except:
```

```

    ), (
        data.user_name, data.user_email, data.age, data.gender,
        data.education, data.occupation, data.sleep_pattern, data.stress_source,
        data.mental_history, data.support_system, data.stressful_events,
        data.phq_little_interest, data.phq_feeling_down,
        data.phq_sleep_issues, data.phq_low_energy, data.phq_appetite,
        data.phq_self_worth, data.phq_concentration,
        data.phq_psychomotor, data.phq_suicidal_thoughts,
        data.phq_total_score,
        data.gad_nervous, data.gad_uncontrollable_worry,
        data.gad_excessive_worry, data.gad_trouble_relaxing,
        data.gad_restlessness, data.gad_irritability, data.gad_fear,
        data.gad_total_score,
        dep_pred, anx_pred, round(float(confidence), 2), escalation["risk_level"],
        mhi["mhi_score"], mhi["mhi_label"],
        escalation["escalation_score"], warning["warning_level"],
        datetime.now(),
    ))
    conn.commit()
    cur.close()
    conn.close()
except Exception as e:
    print(f"⚠️ DB Insert Error: {e}")

return {
    "depression_severity": dep_pred,
    "anxiety_severity": anx_pred,
    "confidence_percentage": confidence,
    "top_depression_factors": top_dep_features,
    "top_anxiety_factors": top_anx_features,
    "all_depression_shap": all_dep_shap,
    "all_anxiety_shap": all_anx_shap,
    "mental_health_index": mhi,
    "early_warning": warning,
    "risk_escalation": escalation,
    "ai_recommendation": ai_text,
}

```

# APPENDIX B

## PAPER SUBMISSION AND ACCEPTANCE PROOF

### Submission:

#### Manuscript Submission for ICICV 2026 External



**ABHISHEK SRIRAM (PA2312052010054)** <as4484@srmist.edu.in>  
to icicv.conf, Ishwarya ▾

Mar 17, 2026, 5:40 PM ☆ 😊 ↩ ⋮

Dear ICICV 2026 Organizing Committee,

I am pleased to submit our research manuscript entitled "**MindPulse: A Multimodal AI Framework for Early Detection of Mental Health Crisis in Young Adults**" for consideration at the **7th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV 2026)**.

The paper proposes a novel AI-driven framework that integrates **machine learning-based risk prediction, explainable artificial intelligence (SHAP), temporal modeling for longitudinal mental health monitoring, and large language model-based personalized recommendations**. The system is designed as a real-time deployable architecture implemented using FastAPI and React, enabling interpretable risk prediction and early intervention for mental health crises.

The research aligns well with the conference themes related to **AI-driven computing applications, contextual awareness, multimodal systems, and explainable AI**. We believe the proposed framework contributes to advancing intelligent healthcare technologies by combining predictive modeling with interpretable decision support and real-time deployment capabilities.

Please find the **manuscript attached in IEEE conference format** for your review and consideration.

#### Paper Information:

**Title:** MindPulse: A Multimodal AI Framework for Early Detection of Mental Health Crisis in Young Adults

**Author:** Abhishek Sriram

**Affiliation:** Department of Data Science and Business Systems

School of Computing

SRM Institute of Science and Technology

Kattankulathur 603203

**Email:** [as4484@srmist.edu.in](mailto:as4484@srmist.edu.in)

**Author:** Ishwarya K

**Affiliation:** Department of Computing Technologies

School of Computing

SRM Institute Of Science and Technology

Kattankulathur 603203

**Email:** [ishwaryak3@srmist.edu.in](mailto:ishwaryak3@srmist.edu.in)

Kindly review our manuscript and let us know if any additional documents or submission procedures are required.

We appreciate your time and consideration and look forward to the opportunity to present our work at ICICV 2026.

Thank you.

Sincerely,

Abhishek Sriram

Department of Data Science and Business Systems

School of Computing

SRM Institute of Science and Technology

Kattankulathur 603203

Email: [as4484@srmist.edu.in](mailto:as4484@srmist.edu.in)

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**Acceptance:**

|                                                                                                                                                                   |                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|                                                                                  | <b>7<sup>th</sup> International Conference on<br/>Intelligent Communication Technologies and Virtual Mobile Networks</b> |
| <div style="background-color: #0056b3; color: white; padding: 2px 10px; border-radius: 10px; display: inline-block;"> <b>20 - 22 May 2026 - ICICV 2026</b> </div> |                                                                                                                          |
| <a href="https://icicv.org/2026/">https://icicv.org/2026/</a>   <a href="mailto:icicv.conf@gmail.com">icicv.conf@gmail.com</a>                                    |                                                                                                                          |

***Letter of Acceptance*****Paper ID:** ICICV-1230**Paper Title:** MindPulse: A Multimodal AI Framework for Early Detection of Mental Health Crisis in Young Adults**Author(s) Name:** Abhishek Sriram, Ishwarya K**Affiliation Details:** SRM Institute of Science and Technology**Dear Author(s),**

It is with great pleasure that we extend our warmest congratulations to you on the acceptance of the above-mentioned paper for oral presentation and publication in the 7<sup>th</sup> International Conference on Intelligent Communication Technologies and Virtual Mobile Networks, scheduled to be held in Francis Xavier Engineering College, Tirunelveli, India from May 20<sup>th</sup> to 23<sup>th</sup>, 2026.

Your submission was subjected to a rigorous review process, and the result that your paper has been selected for inclusion in our conference program. We believe that your contribution will greatly enrich the discussions and knowledge exchange at our event. Your participation will undoubtedly contribute to the success of the 7<sup>th</sup> International Conference on Intelligent Communication Technologies and Virtual Mobile Networks.

Once again, congratulations on your acceptance, and we anticipate your valuable contribution to our conference.

Best Regards,




Conference Chair - ICICV 2026

# APPENDIX C

## PLAGIARISM REPORT



Page 2 of 74 - Integrity Overview

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